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Diehl et al.

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(54) **ANTENNA FOR A MOTORIZED WINDOW TREATMENT**

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(51) **Int. Cl.**
E06B 9/72 (2006.01)
E06B 9/68 (2006.01)

(52) **U.S. Cl.**
CPC ***E06B 9/72*** (2013.01); *E06B 2009/6809* (2013.01)

(58) **Field of Classification Search**
CPC E06B 9/42; E06B 9/50; E06B 9/72; E06B 2009/6809; E06B 9/44; E06B 9/40; E06B 9/68
See application file for complete search history.

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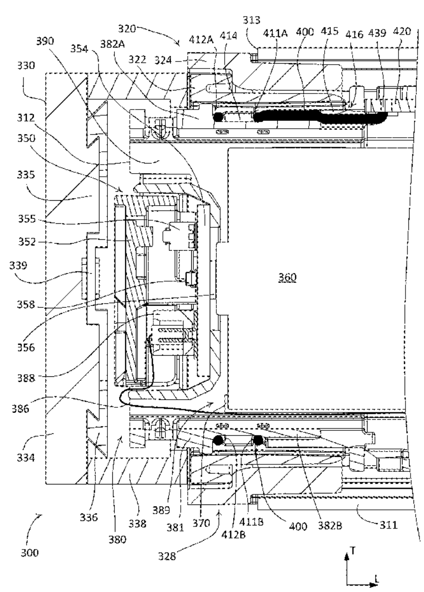
Primary Examiner — Abe Massad
(74) *Attorney, Agent, or Firm* — Flaster Greenberg P.C.

(57) **ABSTRACT**

A motorized window treatment may comprise an antenna that allows for wireless communication. The motorized window treatment may comprise a roller tube, a motor drive unit, and at least one mounting bracket. The roller tube may be configured to windingly receive a flexible material and to be rotated to raise and lower the flexible material. The mounting bracket may be configured to support a bearing assembly of the motor drive unit to allow the roller tube to rotate with respect to the mounting bracket. The bearing assembly may be located between the roller tube and the mounting bracket, so as to form a gap between the roller tube and the mounting bracket. The antenna may comprise an electrical conductor wrapped around the motor drive unit adjacent to the gap between the roller tube and the mounting bracket.

23 Claims, 16 Drawing Sheets



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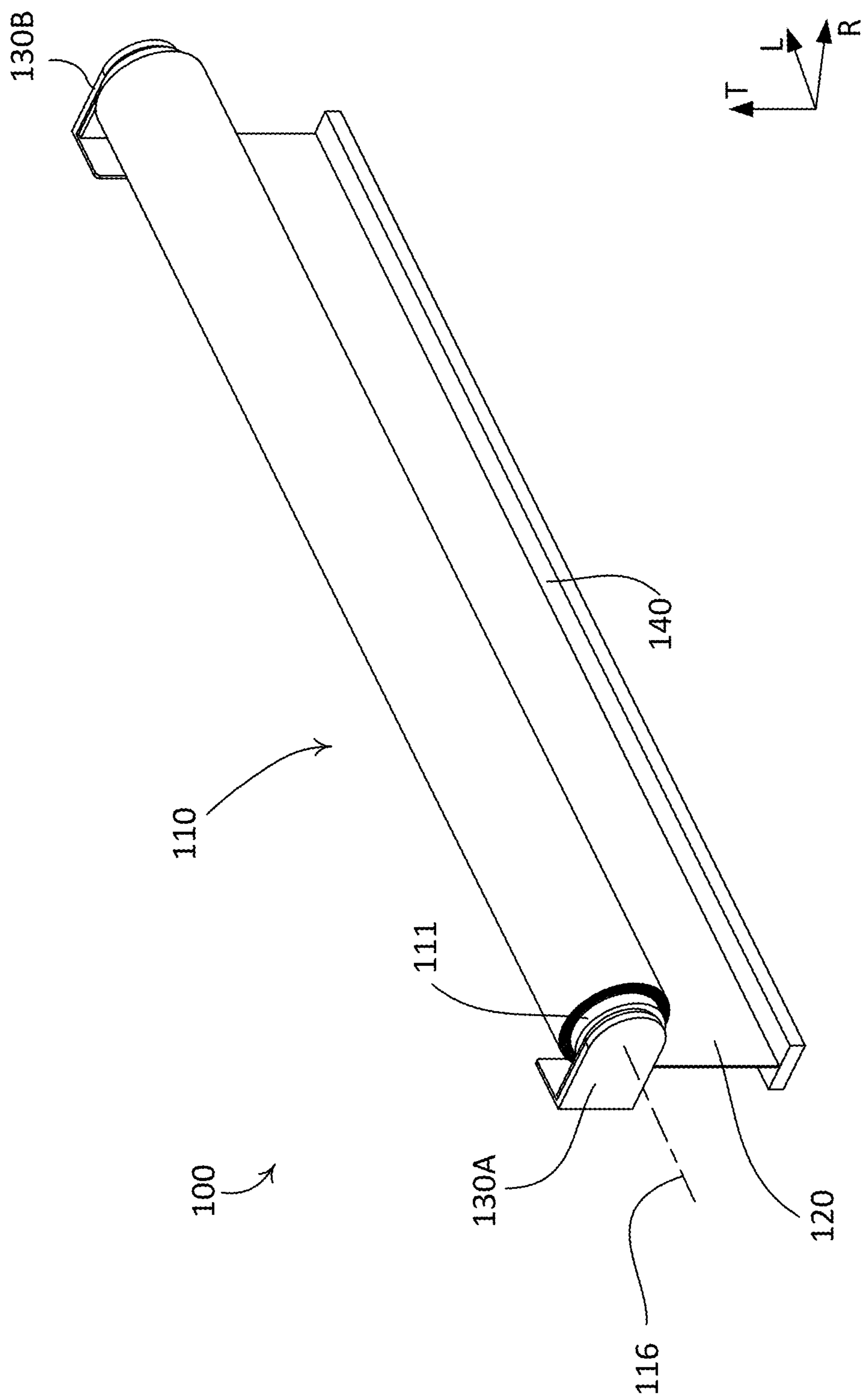
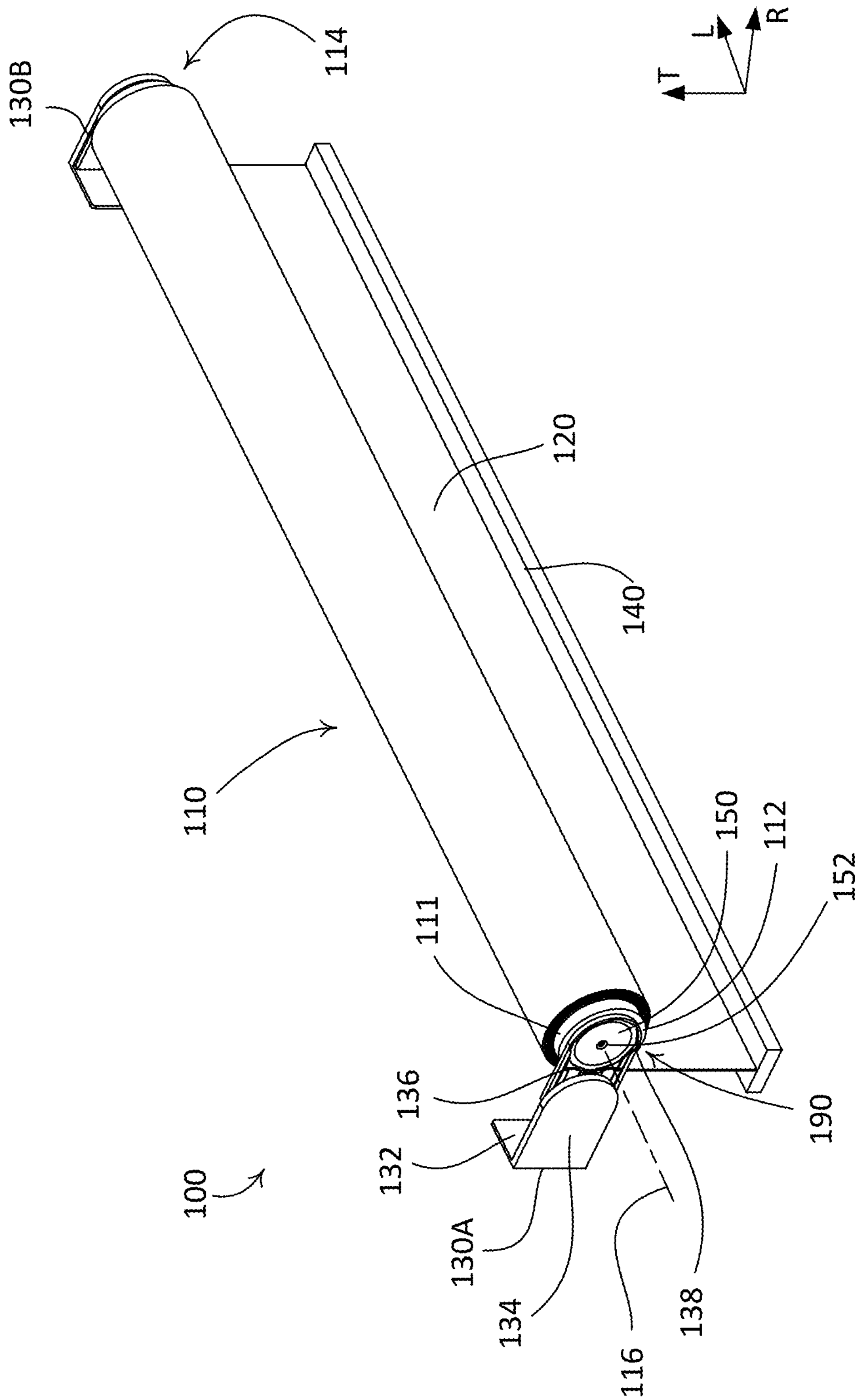
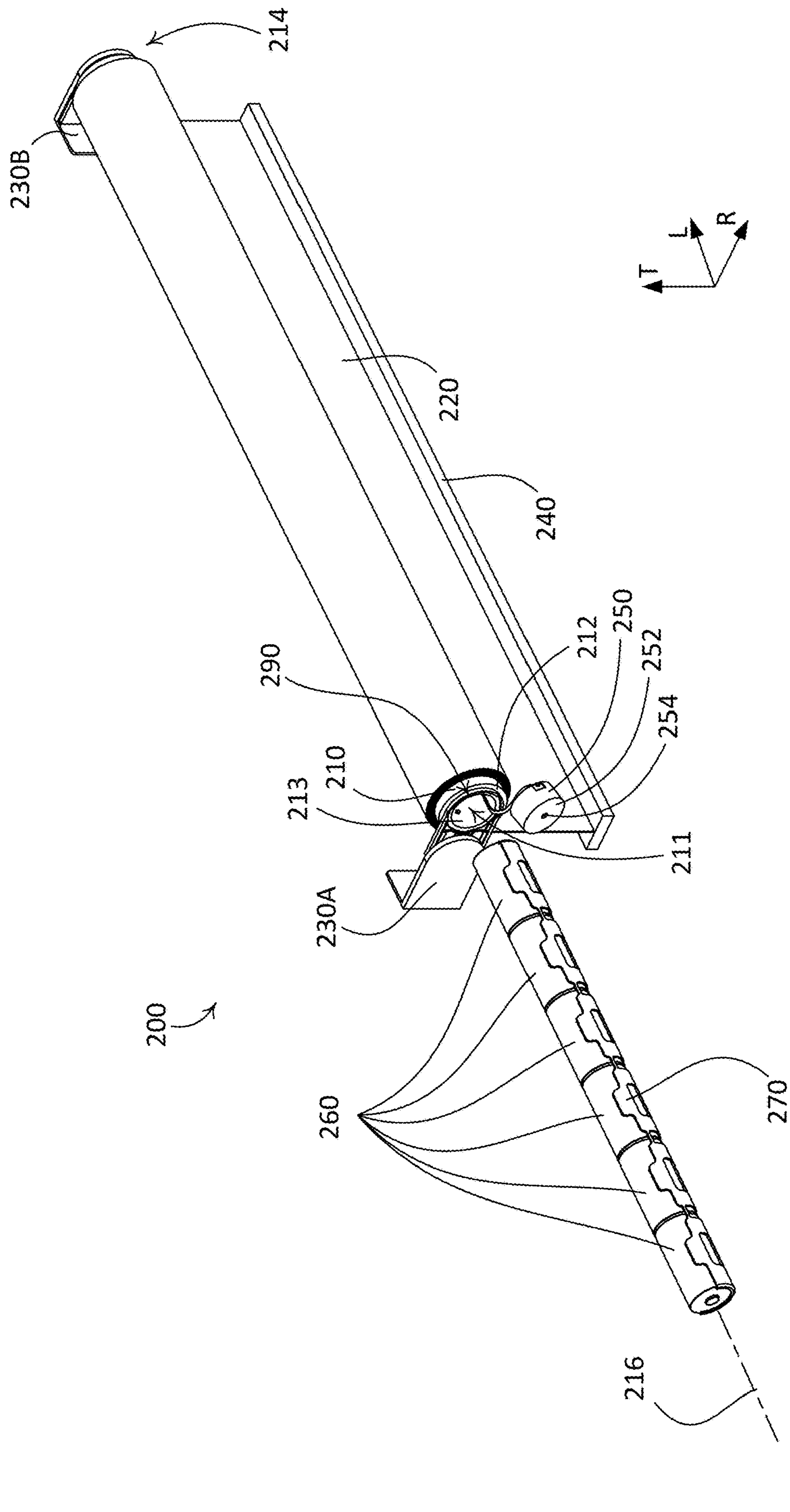


FIG. 1



114
130B
T
L
R
120
140
110
150
112
152
111
136
190
132
100
130A
134
138
116

FIG. 2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230B
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290
212
250
252
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210
213
211
230A
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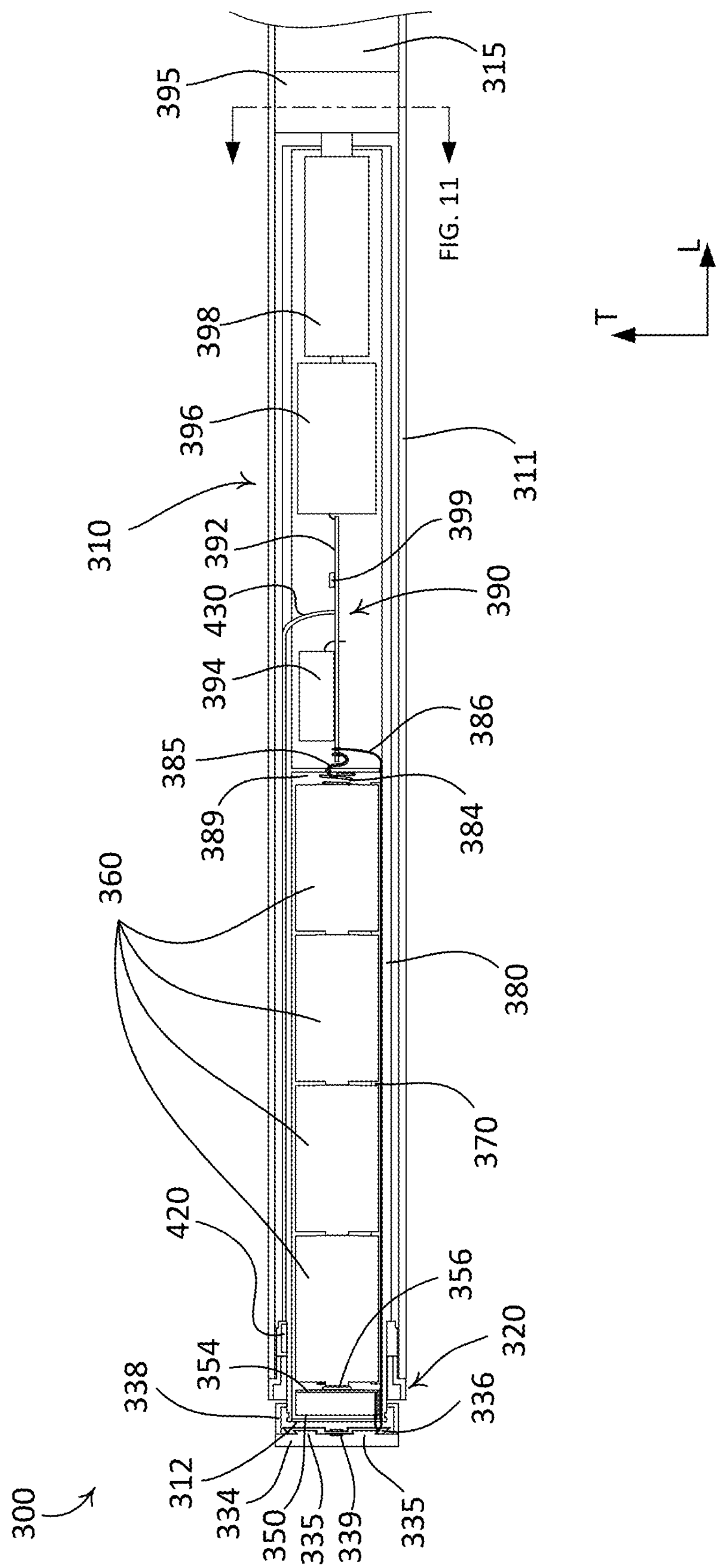
260
270
216

FIG. 3


315
395
FIG. 11
L
T
398
396
310
311
392
399
430
390
394
386
385
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384
360
380
370
420
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354
320
338
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312
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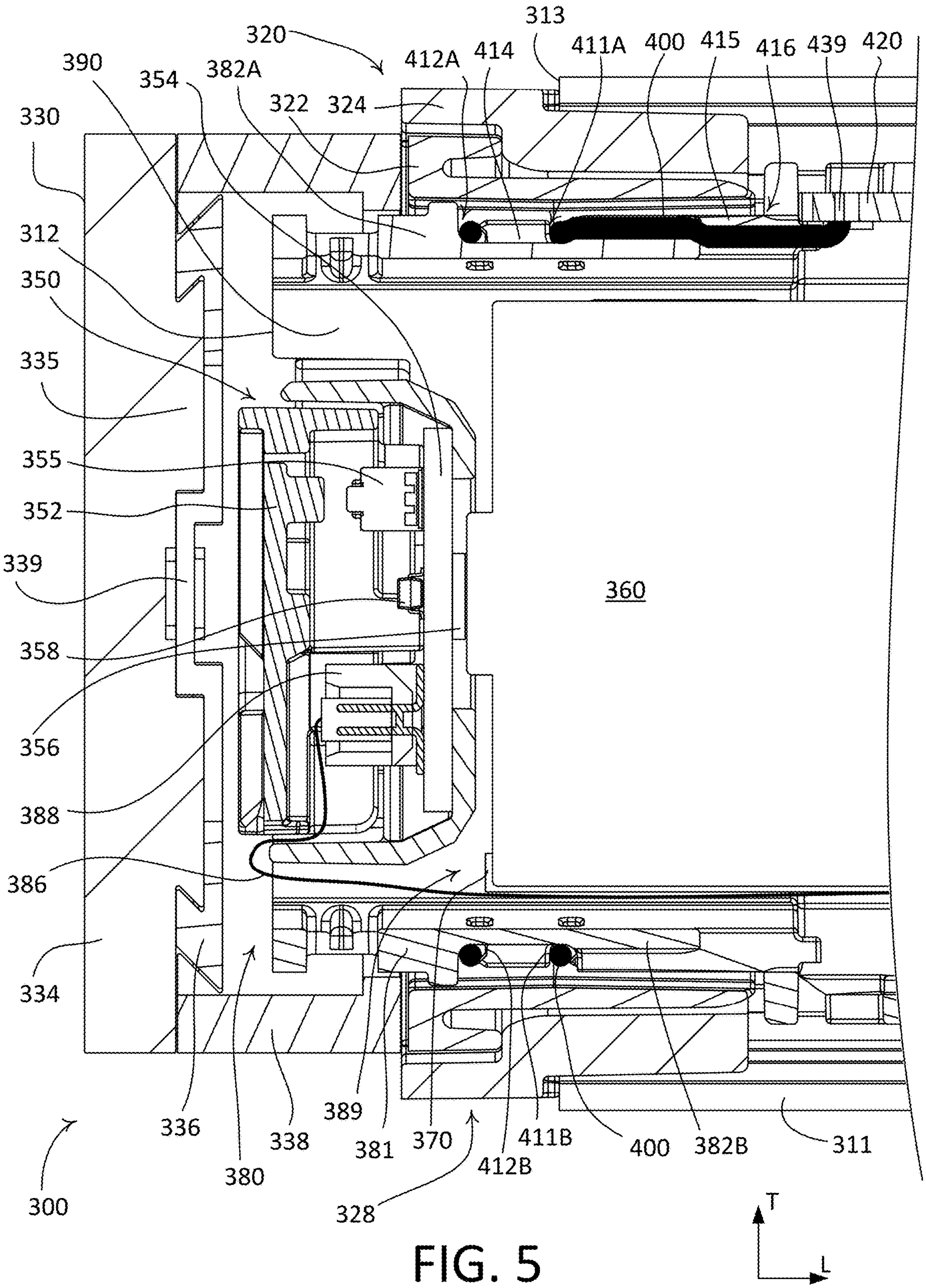
339
335
300

FIG. 4


313
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412A
414
411A
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415
416
439
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382A
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352
339
360
358
356
388
386
334
336
338
389
381
370
412B
411B
400
382B
311
380
300
328
T
L

FIG. 5

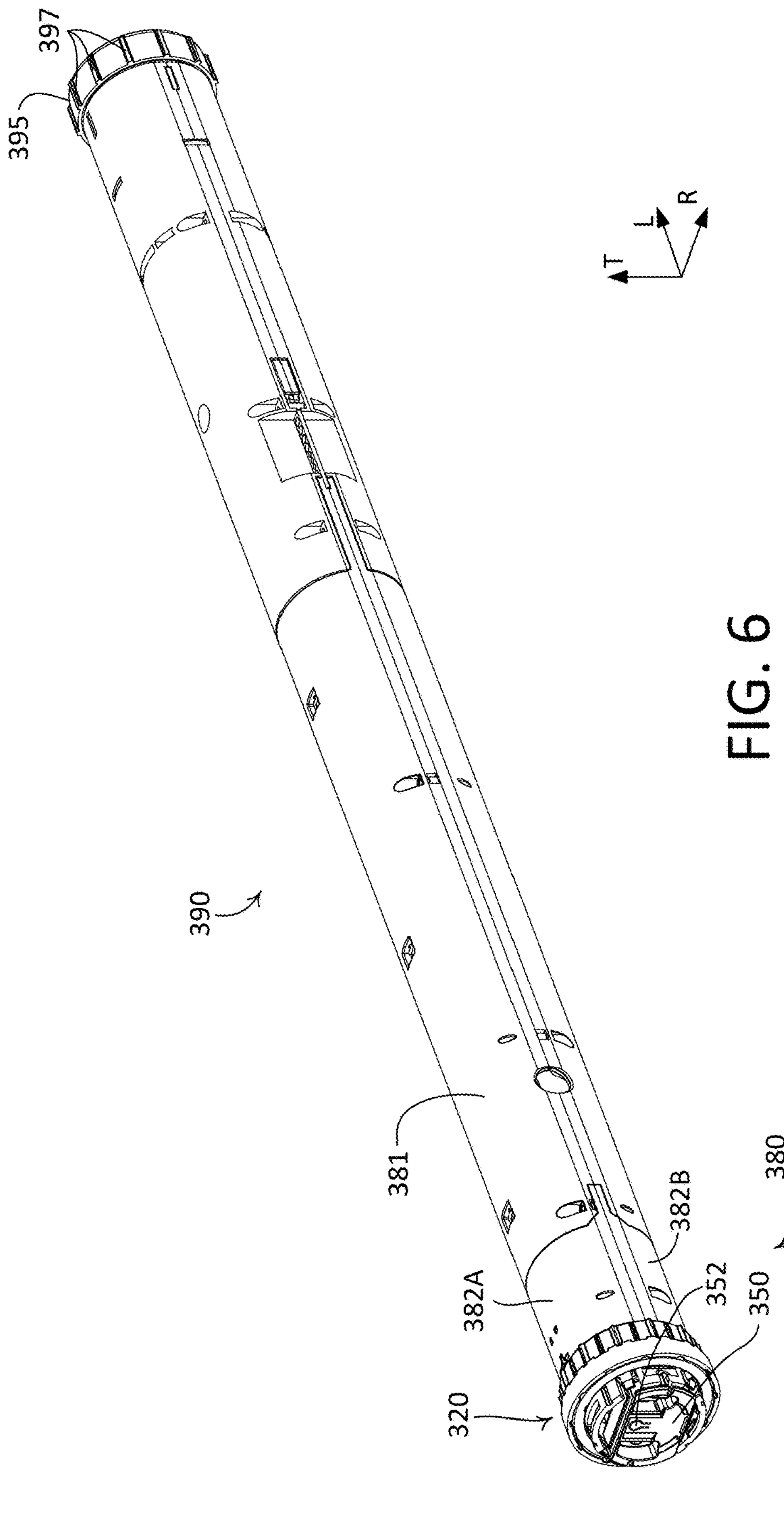


FIG. 6

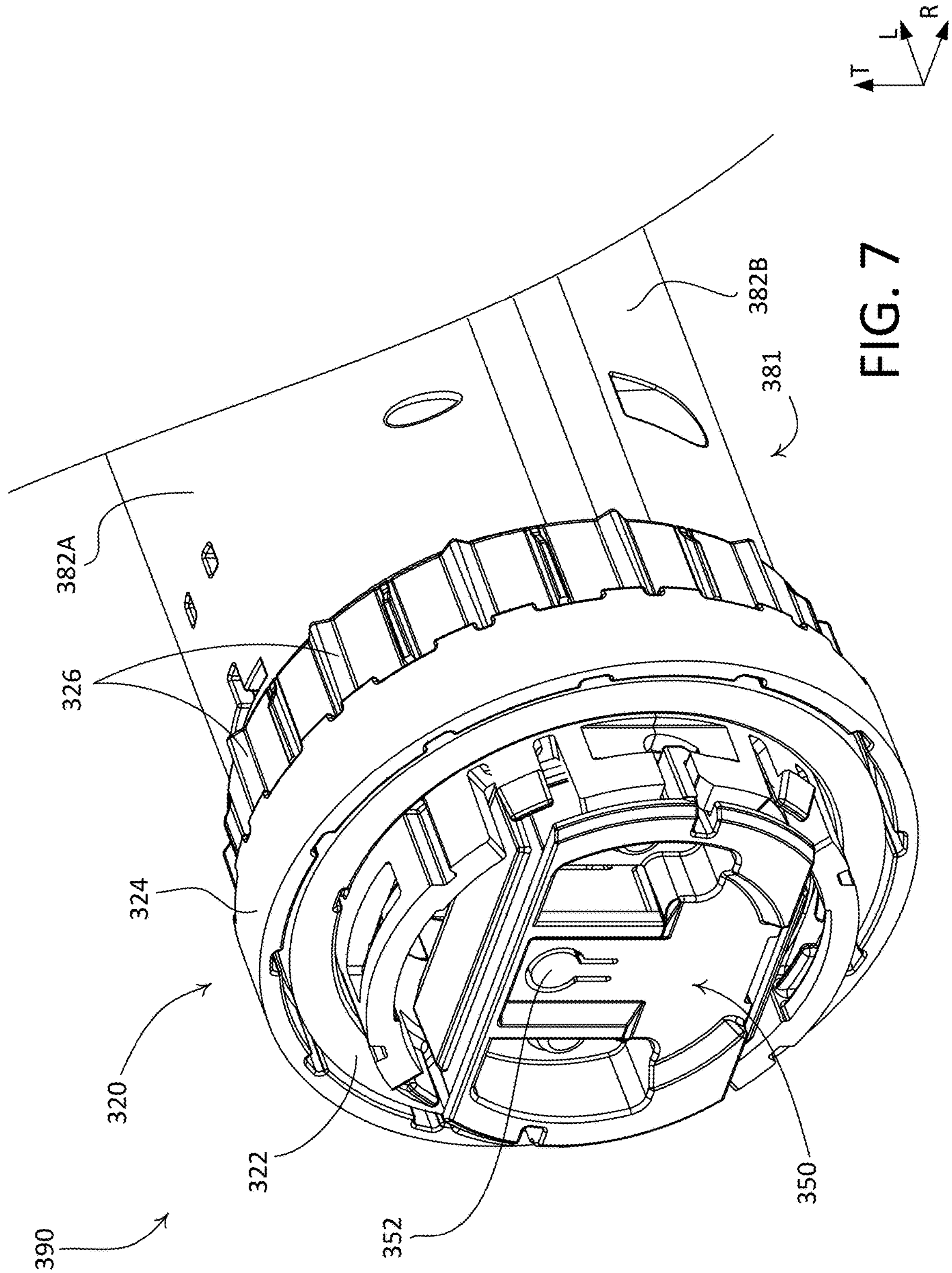
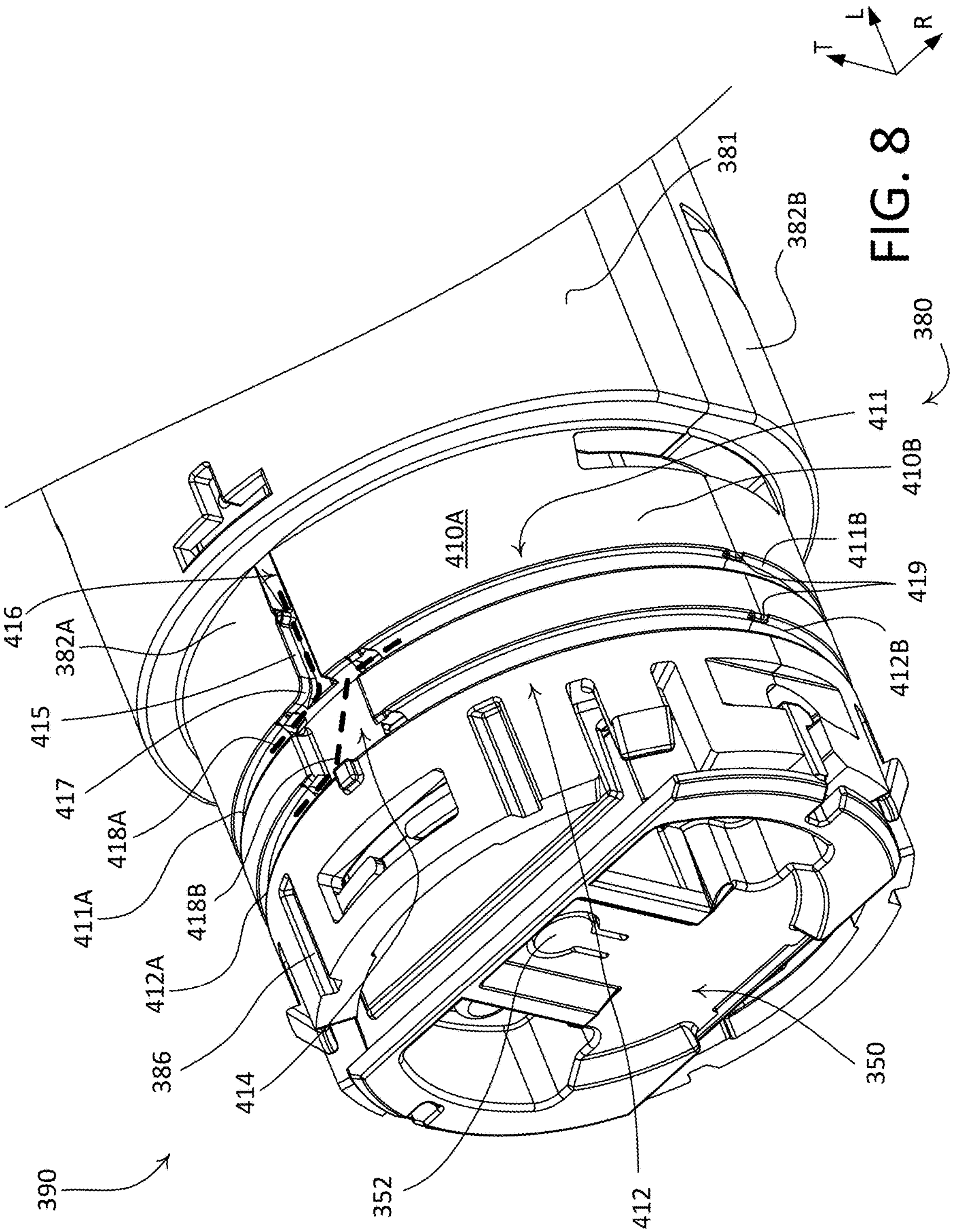


FIG. 7


L
T
R
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382B
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410A
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412B
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382A
415
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411A
418B
412A
386
414
350
390
352
412

FIG. 8

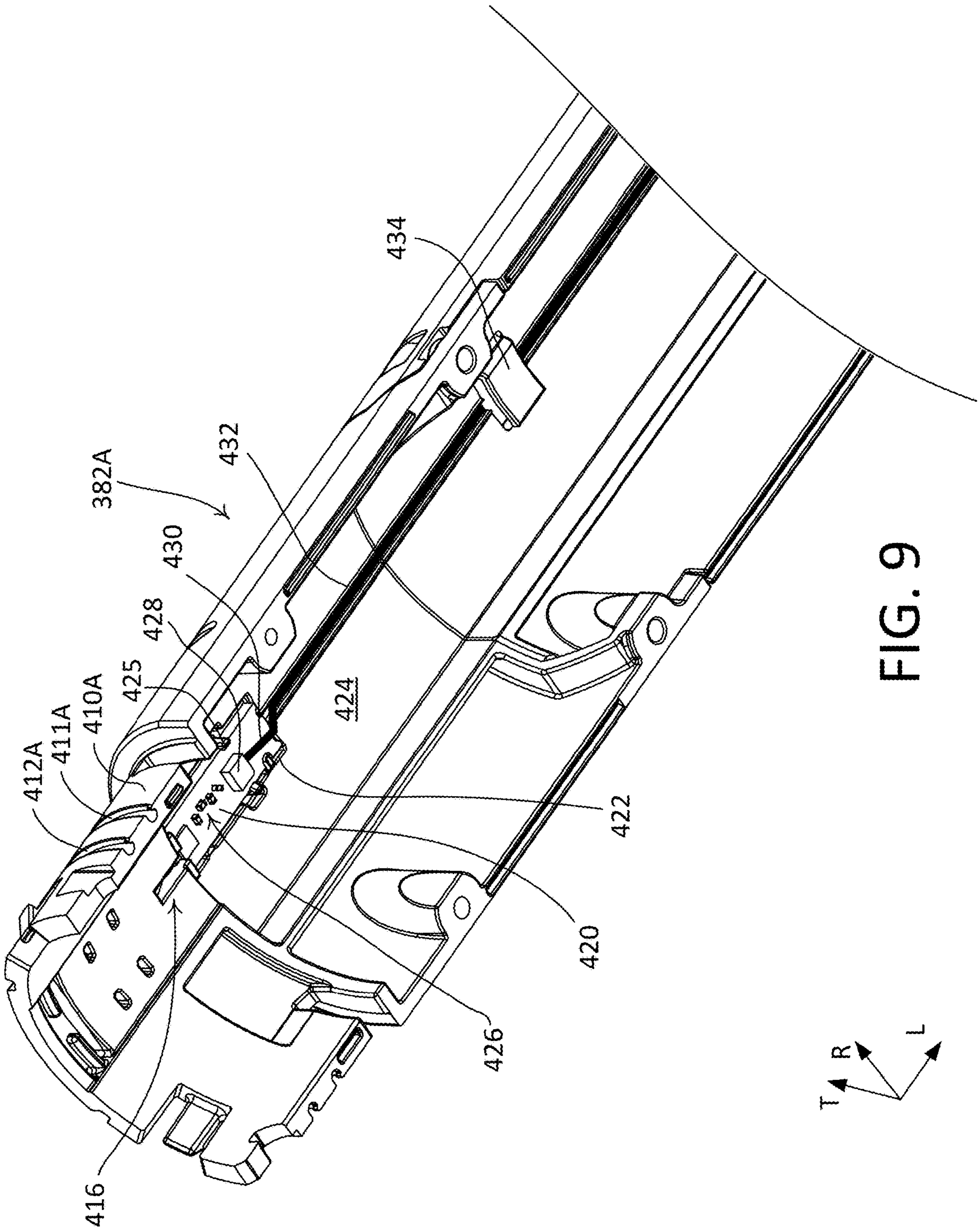
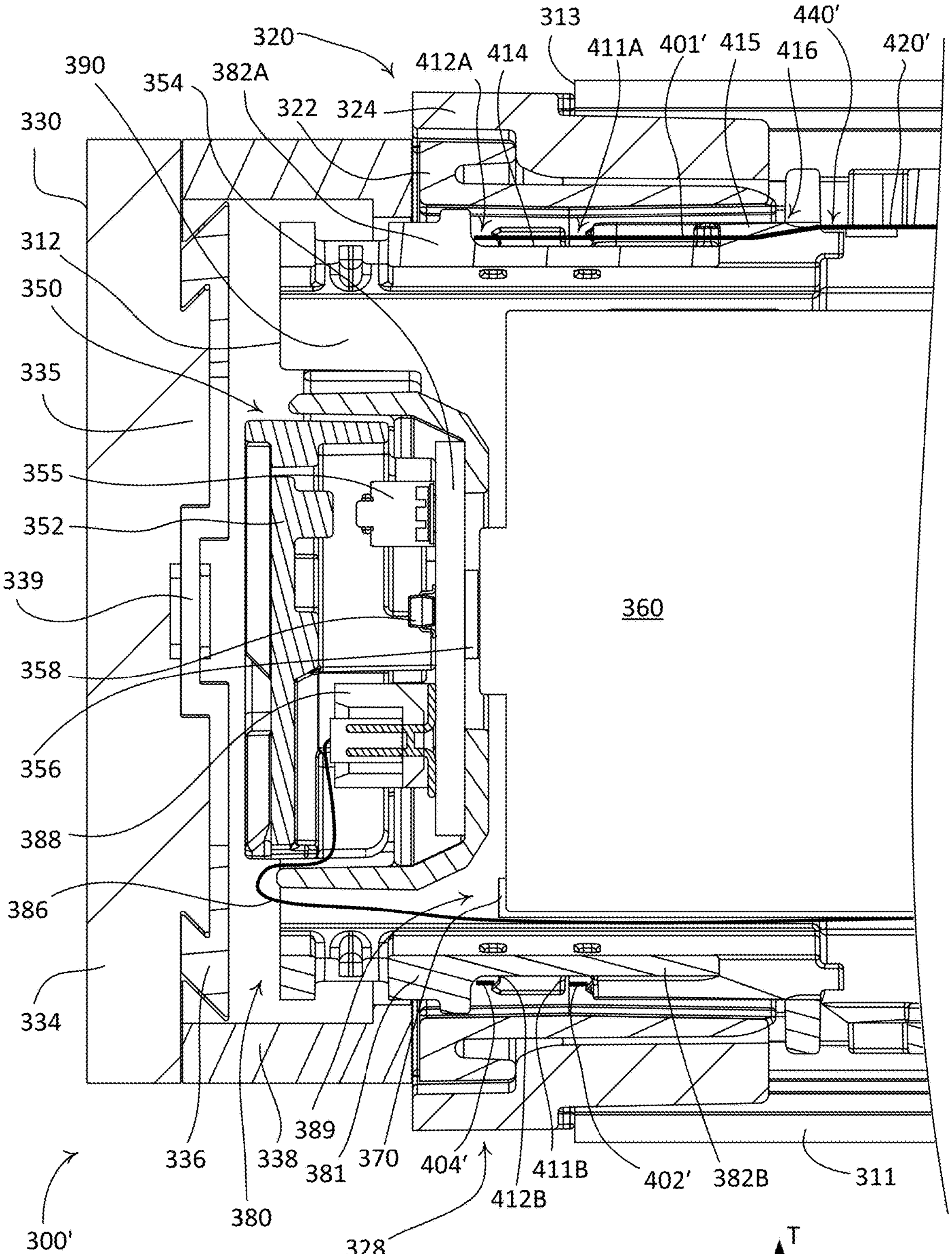


FIG. 9



320
313
440′
390
354
382A
412A
414
411A
401′
415
416
420′
330
322
324
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335
355
352
339
360
358
356
388
386
334
336
338
389
381
370
404′
411B
402′
382B
311
380
412B
300'
328
T
L

FIG. 10

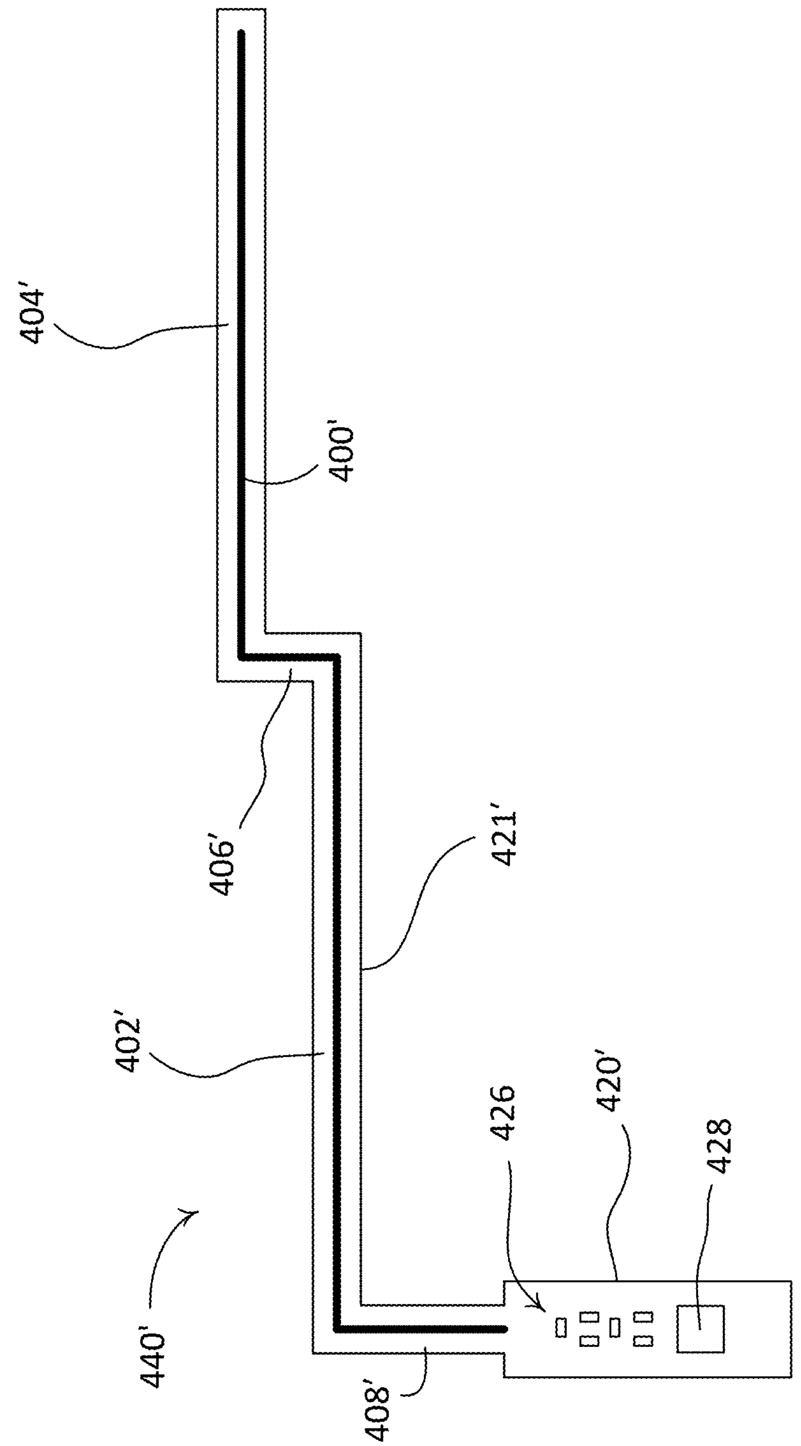
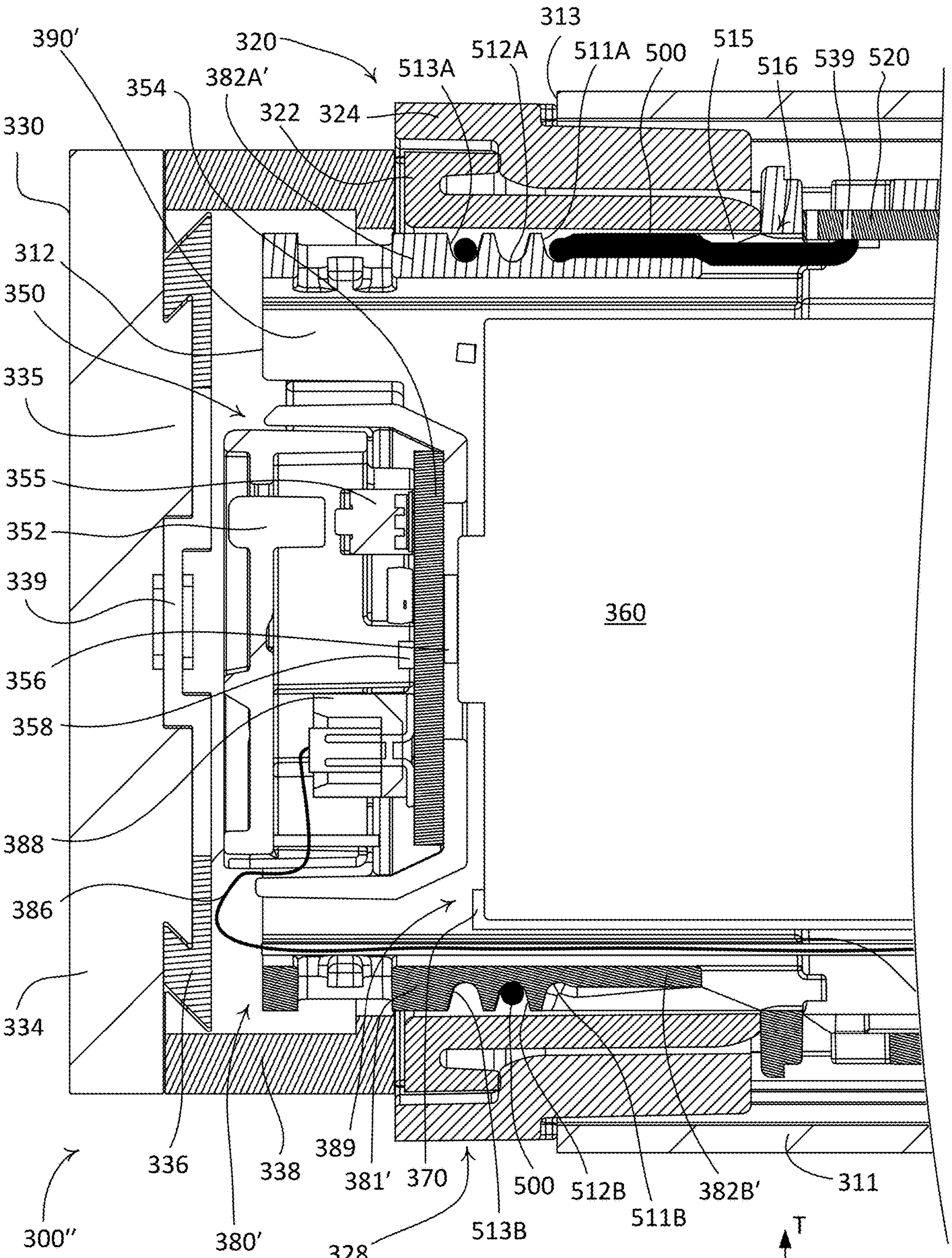


FIG. 11


390′
320
313
513A
512A
511A
500
515
516
539
520
354
382A′
322
324
330
312
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335
355
352
339
356
358
388
386
334
360
336
338
389
381′
370
500
512B
382B′
311
300″
380′
513B
511B
328
T
L

FIG. 12

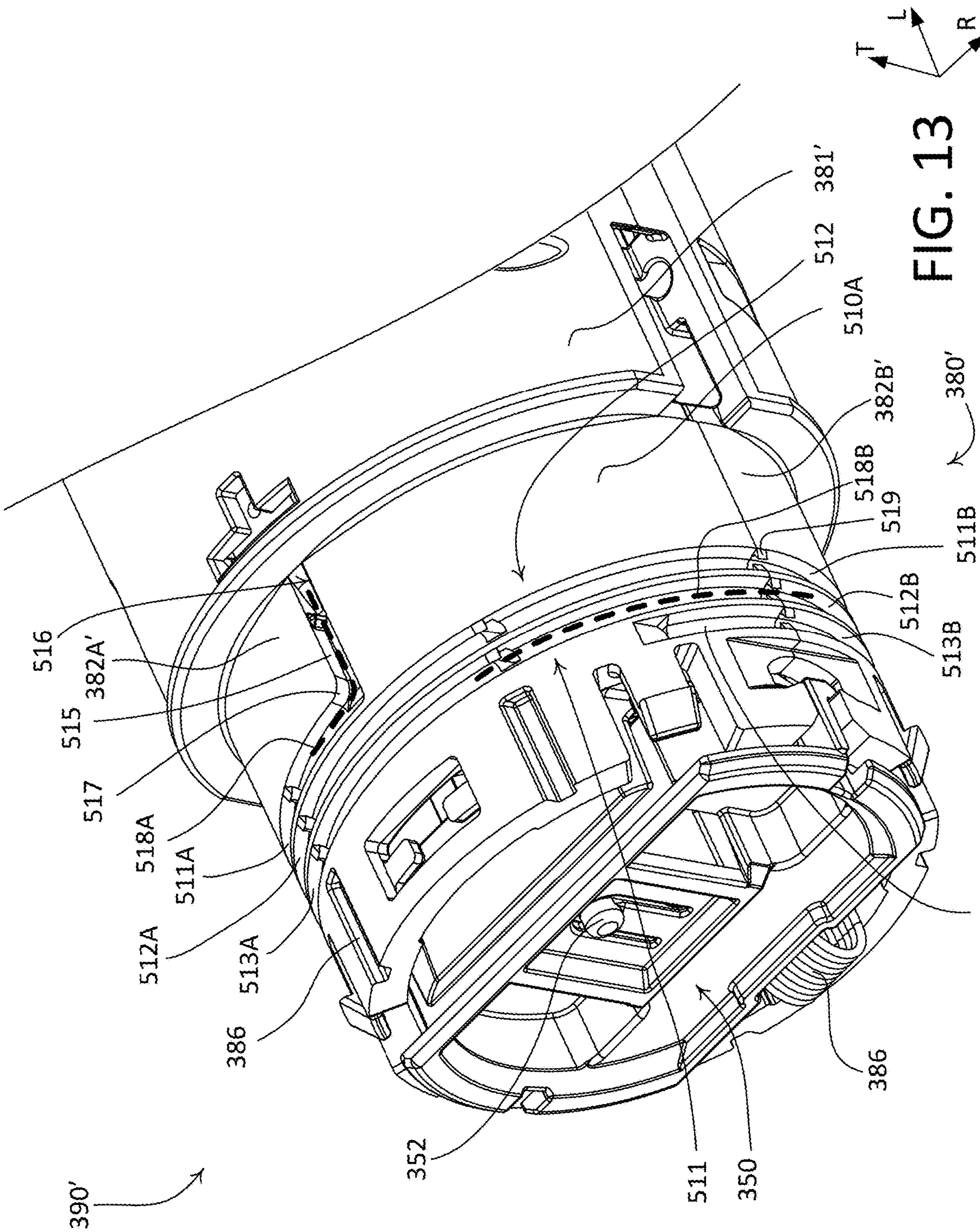
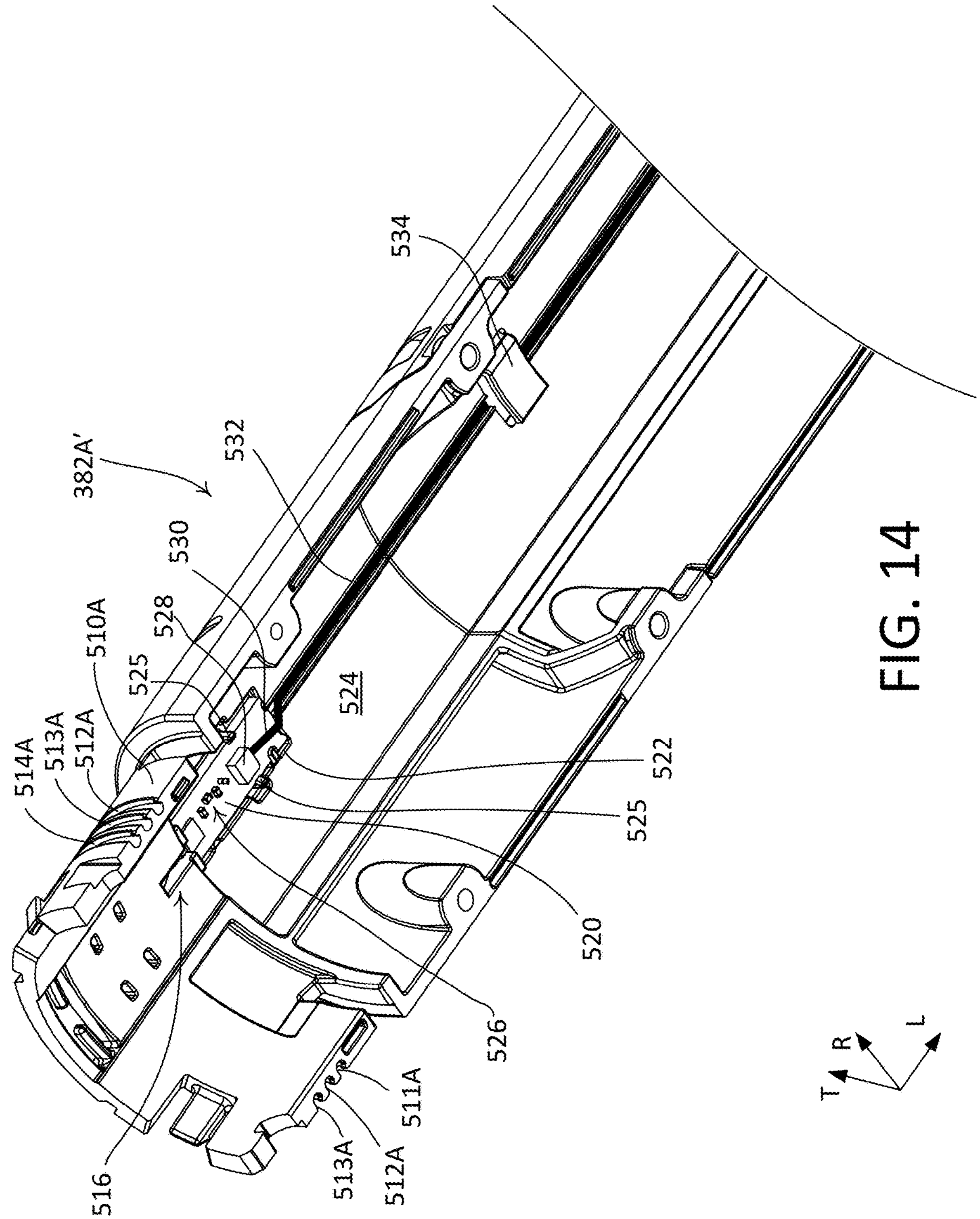

T
L
R
381'
512
510A
382B'
380'
518B
519
511B
512B
513B
514A
516
382A'
515
517
518A
511A
512A
513A
386
352
511
350
386
390'

FIG. 13


534
532
382A'
530
528
510A
525
514A
513A
512A
524
522
525
520
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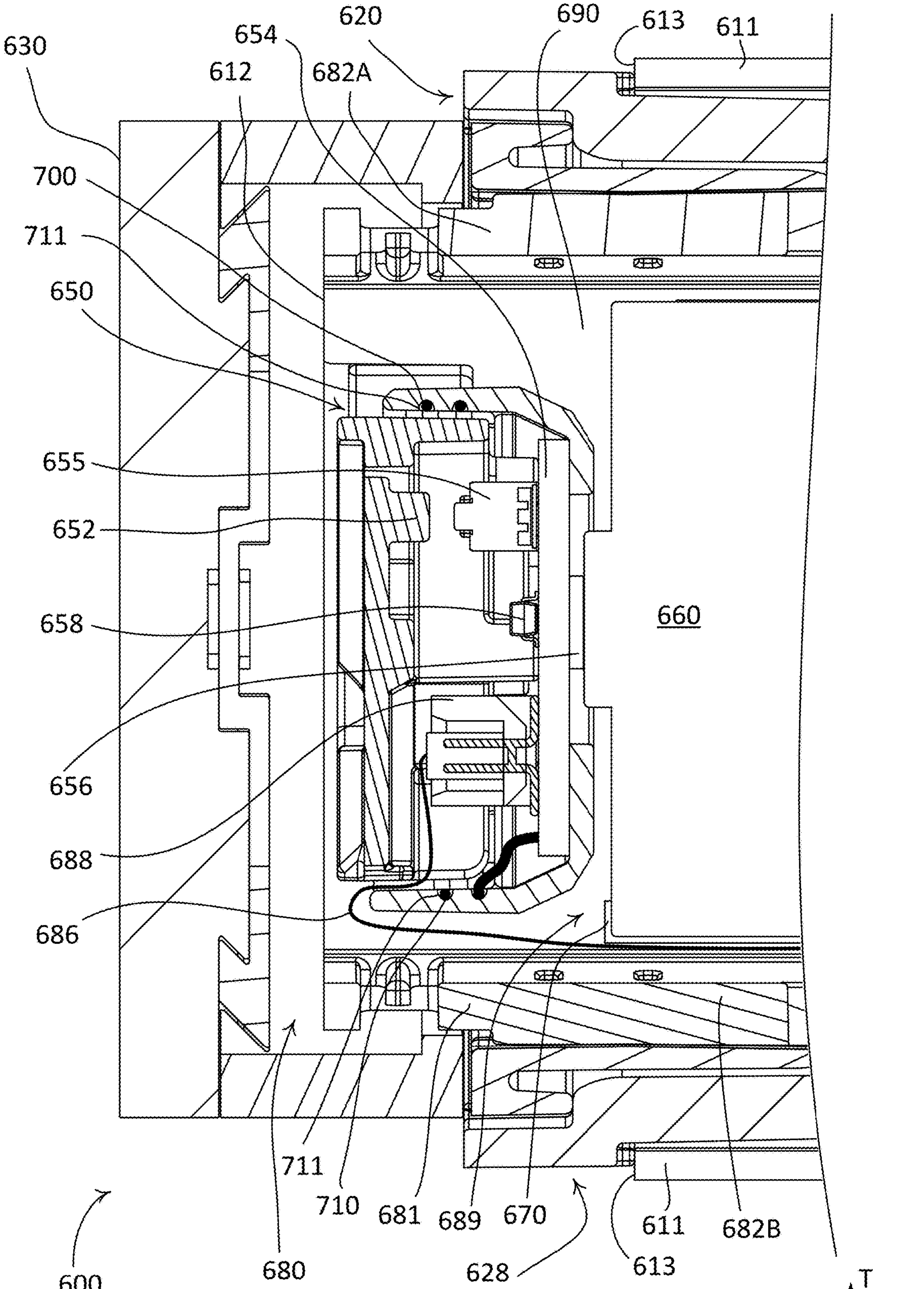
516
513A
512A
511A
T
R
L

FIG. 14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682A
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711
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655
652
658
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686
711
710
681
689
670
611
682B
680
628
613
600
T
L

FIG. 15

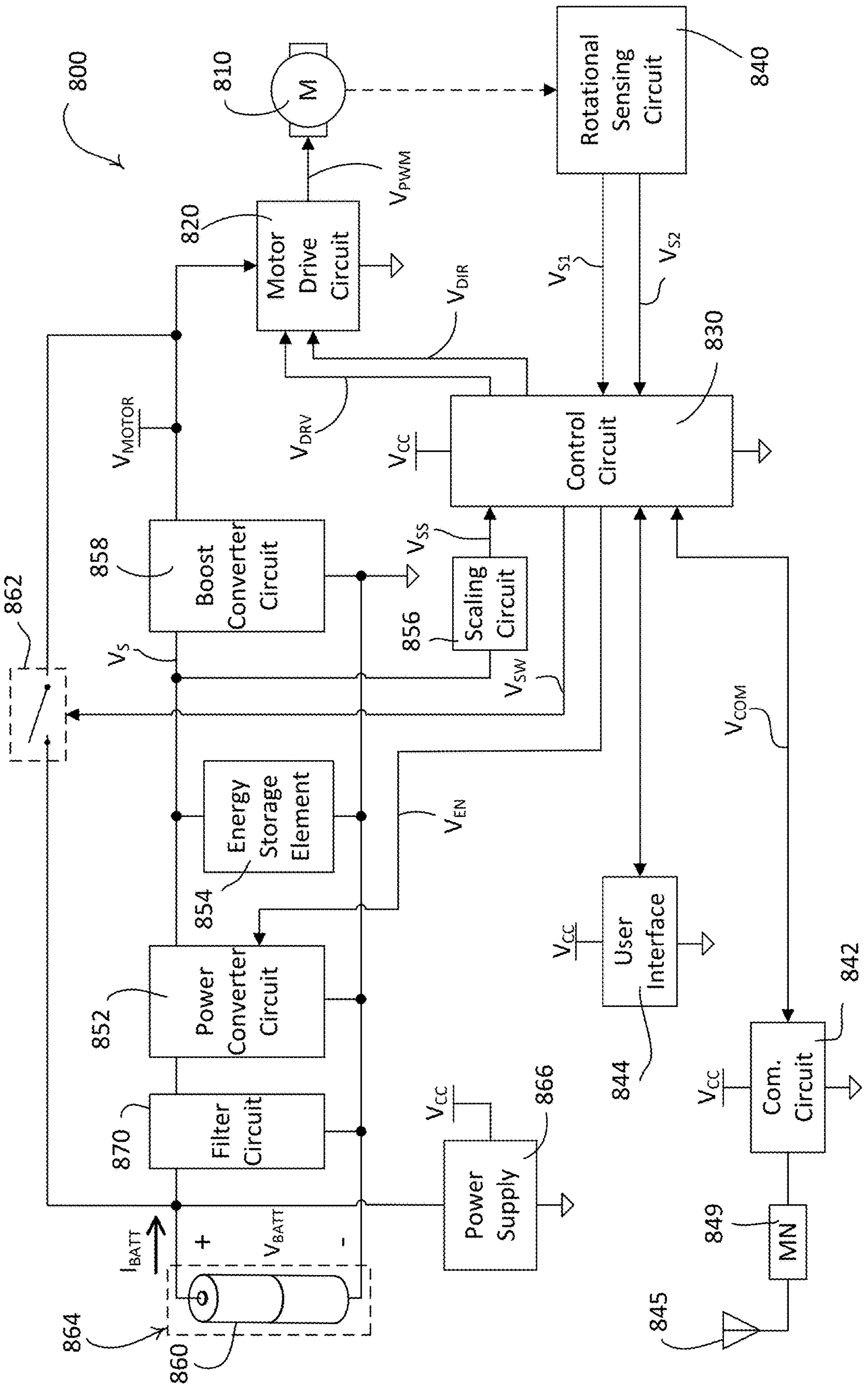
800
810
M
820
Motor Drive Circuit
V_{PWM}
Rotational Sensing Circuit
840
V_{S1}
V_{S2}
830
Control Circuit
V_{DIR}
V_{DRV}
V_{CC}
V_{MOTOR}
858
Boost Converter Circuit
V_{SS}
856
Scaling Circuit
862
V_S
V_{SW}
Energy Storage Element
854
V_{EN}
User Interface
844
V_{CC}
V_{COM}
852
Power Converter Circuit
842
Com. Circuit
V_{CC}
870
Filter Circuit
V_{CC}
866
Power Supply
849
MN
845
I_{BATT}
+
V_{BATT}
-
864
860

FIG. 16

ANTENNA FOR A MOTORIZED WINDOW TREATMENT

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to U.S. provisional patent application No. 63/193,433, filed May 26, 2021, and U.S. provisional patent application No. 63/227,252, filed Jul. 29, 2021, which are incorporated herein by reference in their entirety.

BACKGROUND

A window treatment may be mounted in front of one or more windows, for example to prevent sunlight from entering a space and/or to provide privacy. Window treatments may include, for example, roller shades, roman shades, venetian blinds, or draperies. A roller shade typically includes a flexible shade fabric wound onto an elongated roller tube. Such a roller shade may include a weighted hembar located at a lower end of the shade fabric. The hembar may cause the shade fabric to hang in front of one or more windows over which the roller shade is mounted.

A typical window treatment can be mounted to structure surrounding a window, such as a window frame. Such a window treatment may include brackets at opposed ends thereof. The brackets may be configured to operably support the roller tube, such that the flexible material may be raised and lowered. For example, the brackets may be configured to support respective ends of the roller tube. The brackets may be attached to structure, such as a wall, ceiling, window frame, or other structure.

Such a window treatment may be motorized. A motorized window treatment may include a roller tube, a motor, brackets, and electrical wiring. The components of the motorized window treatment, such as the brackets, the roller tube, electrical wiring, etc. may be concealed by a fascia or installed in a pocket out of view.

SUMMARY

As described herein, a motorized window treatment may comprise an antenna that allows for wireless communication. The motorized window treatment may comprise a roller tube, a motor drive unit, and at least one mounting bracket. The roller tube may be configured to windingly receive a flexible material and to be rotated to raise and lower the flexible material. The motor drive unit may be received within a cavity of the roller tube. The motor drive unit may comprise a motor configured to rotate the roller tube and a bearing assembly coupled to the roller tube, such that the roller tube is configured to rotate around the motor drive unit. The mounting bracket may be configured to support the bearing assembly of the motor drive unit to allow the roller tube to rotate with respect to the mounting bracket. The bearing assembly may be located between the roller tube and the mounting bracket, so as to form a gap between the roller tube and the mounting bracket. The antenna may comprise an electrical conductor wrapped around the motor drive unit adjacent to the gap between the roller tube and the mounting bracket. For example, the antenna may be configured to be spirally wound around the motor drive unit proximate to the gap between the roller tube and the mounting bracket.

A motorized window treatment may comprise a roller tube, a motor drive unit, and at least one mounting bracket. The roller tube may be configured to windingly receive a flexible material. The roller tube may be configured to be rotated to raise and lower the flexible material. The motor drive unit may be received within a cavity of the roller tube. The motor drive unit may comprise a motor, a housing, and an antenna. The motor may be configured to rotate the roller tube. The housing may be configured to house the motor. The housing may comprise at least one channel formed in a surface of the housing. The antenna may comprise an electrical conductor. The at least one mounting bracket may be configured to support the roller tube such that the roller tube can rotate with respect to the at least one mounting bracket. The motorized window treatment may define a gap between the roller tube and the mounting bracket. The electrical conductor of the antenna may be wrapped around the housing of the motor drive unit adjacent to the gap between the roller tube and the mounting bracket. The electrical conductor of the antenna may be configured to be received within the at least one channel when wrapped around the housing.

The motor drive unit may comprise a wireless communication circuit that is electrically coupled to the antenna for transmitting and receiving wireless signals. The at least one channel may comprise at least two peripheral channels that extend parallel to each other around the circumference of the housing in an outer surface of the housing. The at least two peripheral channels may be joined together at a recess and the electrical conductor of the antenna may be configured to pass from one peripheral channel to another via the recess. Alternatively, the at least one channel may comprise a single spiral-shaped channel. The spiral-shaped channel may be configured such that the antenna moves away from the roller tube in the longitudinal direction as the antenna wraps around the housing.

The motor drive unit may comprise a motor drive printed circuit board on which drive circuitry for controlling the motor is mounted. The motor drive unit may comprise a battery compartment for receiving one or more batteries for powering the drive circuitry on the motor drive printed circuit board and the wireless communication circuit. The housing may comprise a cap for covering an end of the battery compartment. The battery compartment may be located between the cap and the motor drive printed circuit board. The electrical conductor of the antenna may be wrapped around the cap. The electrical conductor of the antenna may be located within the at least one channel that extends around the cap. The wireless communication circuit may be located inside the cap.

The motorized window treatment may comprise a matching network circuit that is coupled to the motor drive printed circuit board. The matching network circuit may be located inside the cap. The wireless communication circuit may be coupled to the motor drive printed circuit board via a ribbon cable. The motor drive unit may comprise a coupling printed circuit board located near the gap between the roller tube and the mounting bracket. The coupling printed circuit board may comprise a matching network circuit mounted thereto. The antenna may be electrically coupled to the mounting network circuit on the coupling printed circuit board. The wireless communication circuit may be mounted to the motor drive printed circuit board and electrically connected to the matching network circuit on the coupling printed circuit board by a coaxial cable.

The motorized window treatment may comprise a flexible printed circuit board. The antenna may be formed on the flexible printed circuit board. The motorized window treatment may comprise a bearing assembly coupled to the roller tube, such that the roller tube is configured to rotate around

the motor drive unit. The bearing assembly may be located between the roller tube and the mounting bracket. The bearing assembly may be made of a non-conductive material. The antenna may be wrapped around the housing within an area that surrounds the circumference of the motor drive unit and falls within an area defined by the bearing assembly. At least a portion of the antenna may be aligned with the gap between the roller tube and the at least one mounting bracket. The gap between the roller tube and the at least one mounting bracket may define an area comprising non-conductive components. The roller tube may be made of a conductive material. The antenna may be configured to be electromagnetically coupled to the roller tube. The roller tube and the mounting bracket may both be made of conductive materials. The housing may comprise a body and a cap that is configured to attach to the body. The at least one channel may be defined in the body such that the antenna is wrapped around the body.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an example motorized window treatment.

FIG. **2** is a perspective view of an example battery-powered motorized window treatment with one end of the roller tube in a pivoted position.

FIG. **3** is a perspective view of another example battery-powered motorized window treatment shown with batteries removed.

FIG. **4** is a front cross-sectional view of an example motorized window treatment.

FIG. **5** is an enlarged front cross-sectional view of the motorized window treatment of FIG. **4**.

FIG. **6** is a perspective view of an example motor drive unit of a motorized window treatment, such as the example motorized window treatment of FIG. **4**.

FIG. **7** is an enlarged perspective view of an end portion of the example motor drive unit of FIG. **6**.

FIG. **8** is an enlarged perspective view of the end portion of the example motor drive unit of FIG. **6** with a bearing assembly removed.

FIG. **9** is a partial exploded view of the motor drive unit of FIG. **6** looking up into the inside of an upper portion of a housing of the motor drive unit.

FIG. **10** is an enlarged front cross-sectional view of another example motorized window treatment.

FIG. **11** is a top view of an example flexible printed circuit board having an example antenna.

FIG. **12** is an enlarged front cross-sectional view of another example motorized window treatment.

FIG. **13** is an enlarged perspective view of an end portion of a motor drive unit of the example motorized window treatment of FIG. **12** with a bearing assembly removed.

FIG. **14** is a partial exploded view of the motor drive unit of FIG. **13** looking up into the inside of an upper portion of a housing of the motor drive unit.

FIG. **15** is an enlarged front cross-sectional view of another example motorized window treatment.

FIG. **16** is a block diagram of an example motor drive unit of a motorized window treatment.

DETAILED DESCRIPTION

FIGS. **1** and **2** depict an example motorized window treatment **100** (e.g., a battery-powered motorized window treatment system) that includes a window treatment assembly **110** and one or more mounting brackets **130**A, **130**B. The window treatment assembly **110** may comprise a roller tube **111**, a flexible material **120** (e.g., a covering material) windingly attached to the roller tube **111**, a motor drive unit **190** installed inside of a first end **112** of the roller tube **111**, and an idler (not shown) installed inside of a second end **114** of the roller tube **111**. The mounting brackets **130**A, **130**B may be configured to be coupled to or otherwise mounted to a structure. For example, each of the mounting brackets **130**A, **130**B may be configured to be mounted to (e.g., attached to) a window frame (e.g., to a head jamb or side jambs of the window frame), a wall, a ceiling, or other structure, such that the motorized window treatment **100** is mounted proximate to an opening (e.g., over the opening or in the opening), such as a window for example. The mounting brackets **130**A, **130**B may be configured to be mounted to a vertical structure (e.g., wall-mounted to a wall as shown in FIG. **1**) and/or mounted to a horizontal structure (e.g., ceiling-mounted to a ceiling). For example, the mounting brackets **130**A, **130**B may be rotated **90** degrees from what is shown in FIG. **1**.

The roller tube **111** may operate as a rotational element of the motorized window treatment **100**. The roller tube **111** may be elongate along a longitudinal direction L and rotatably mounted (e.g., rotatably supported) by the mounting brackets **130**. The roller tube **111** may define a longitudinal axis **116**. The longitudinal axis **116** may extend along the longitudinal direction L. The mounting bracket **130**A may extend from the structure in a radial direction R, as shown in FIG. **1**. It should be appreciated that when the mounting brackets **130**A, **130**B are ceiling-mounted, the mounting bracket **130**A may extend from the structure in a transverse direction T. The radial direction R may be defined as a direction perpendicular to the structure and the longitudinal axis **116**. The flexible material **120** may be windingly attached to the roller tube **111**, such that rotation of the roller tube **111** causes the flexible material **120** to wind around or unwind from the roller tube **111** along a transverse direction T that extends perpendicular to the longitudinal direction L. For example, rotation of the roller tube **111** may cause the flexible material **120** to move between a fully-raised position (e.g., an open or fully-open position as shown in FIG. **1**) and a fully-lowered position (e.g., a closed or fully-closed position) along the transverse direction T.

The roller tube **111** may be made of aluminum. The roller tube **111** may be a low-deflection roller tube and may be made of a material that has high strength and low density, such as carbon fiber. The roller tube **111** may have, for example, a diameter of approximately two inches. For example, the roller tube **111** may exhibit a deflection of less than ¼ of an inch when the flexible material **120** has a length of 12 feet and a width of 12 feet (e.g., and the roller tube **111** has a corresponding width of 12 feet and the diameter is two inches). Examples of low-deflection roller tubes are described in greater detail in U.S. Patent Application Publication No. 2016/0326801, published Nov. 10, 2016, entitled LOW-DEFLECTION ROLLER SHADE TUBE FOR LARGE OPENINGS, the entire disclosure of which is hereby incorporated by reference.

The flexible material **120** may include a first end (e.g., a top or upper end) that is coupled to the roller tube **111** and a second end (e.g., a bottom or lower end) that is coupled to a hembar **140**. The hembar **140** may be configured, for example weighted, to cause the flexible material **120** to hang vertically. Rotation of the roller tube **111** may cause the hembar **140** to move toward or away from the roller tube **111** between the raised and lowered positions.

The flexible material **120** may be any suitable material, or form any combination of materials. For example, the flexible material **120** may be "scrim," woven cloth, non-woven material, light-control film, screen, and/or mesh. The motorized window treatment **100** may be any type of window treatment. For example, the motorized window treatment **100** may be a roller shade as illustrated, a soft sheer shade, a drapery, a cellular shade, a Roman shade, or a Venetian blind. As shown, the flexible material **120** may be a material suitable for use as a shade fabric, and may be alternatively referred to as a flexible material. The flexible material **120** is not limited to shade fabric. For example, in accordance with an alternative implementation of the motorized window treatment **100** as a retractable projection screen, the flexible material **120** may be a material suitable for displaying images projected onto the flexible material **120**.

The motorized window treatment **100** may include a motor drive unit (e.g., a drive assembly) that may at least partially be disposed within the roller tube **111**. For example, the motor drive unit may include a housing that is received within the roller tube **111**. The motor drive unit may comprise a motor for rotating the roller tube **111** and a control circuit (e.g., that may include a microprocessor) for controlling the motor. The motor drive unit may be powered by a power source (e.g., an alternating-current or direct-current power source) provided by electrical wiring and/or batteries. The motor drive unit may be operably coupled to the roller tube **111** such that when the motor is controlled, the roller tube **111** rotates. The motor drive unit may be configured to rotate the roller tube **111** of the example motorized window treatment **100** such that the flexible material **120** is operable between the fully-raised position and the fully-lowered position. The motor drive unit may be configured to rotate the roller tube **111** while reducing noise generated by the motor drive unit (e.g., noise generated by one or more gear stages of the drive assembly). Examples of motor drive units for motorized window treatments are described in greater detail in commonly-assigned U.S. Pat. No. 6,497,267, issued Dec. 24, 2002, entitled MOTORIZED WINDOW SHADE WITH ULTRAQUIET MOTOR DRIVE AND ESD PROTECTION, and U.S. Pat. No. 9,598,901, issued Mar. 21, 2017, entitled QUIET MOTORIZED WINDOW TREATMENT SYSTEM, the entire disclosures of which are hereby incorporated by reference.

The motorized window treatment **100** may be configured to enable access to one or more ends of the window treatment assembly **110** while remaining secured to the mounting brackets **130**A, **130**B. For example, the motorized window treatment **100** may be adjusted (e.g., pivoted or slid) between an operating position (e.g., as shown in FIG. **1**) to an extended position (e.g., as shown in FIG. **2**) while secured to the mounting brackets **130**A, **130**B. The operating position may be defined as a position in which the window treatment assembly **110** is supported by and aligned with both mounting brackets **130**A, **130**B. The extended position may be defined as a position in which one or more ends of the window treatment assembly **110** are accessible while still attached to the brackets **130**A, **130**B.

When in the extended position, the one or more ends of the window treatment assembly **110** may be accessed, for example, to replace batteries, adjust one or more settings, make an electrical connection, repair one or more components, and/or the like. One or more of the mounting brackets **130**A, **130**B may enable an end of the window treatment assembly **110** to be accessed when the motorized window treatment is in the extended position. For example, the first mounting bracket **130**A may define a base **132** and an arm **134**. The base **132** and the arm **134** may define a stationary portion of the mounting bracket **130**. The mounting bracket **130**A, **130**B may define a translating portion **136**. The translating portion **136** may include an attachment member **138** that is configured to receive the first end **112** of the window treatment assembly **110**. The attachment member **138** may define an aperture. The base **132** may be configured to attach the mounting bracket **130**A to a structure. When the mounting bracket **130**A is attached to a vertical structure, such as a wall, the arm **134** of the mounting bracket **130**A may extend horizontally (e.g., in the radial direction R) from the base **132**.

One end of the window treatment assembly **110** may slide out when the motorized window treatment **100** is in the extended position. For example, one of the mounting brackets (e.g., the mounting bracket **130**A) may be configured to slide out and the other one of the mounting brackets (e.g., the mounting bracket **130**B) may remain stationary when the motorized window treatment **100** (e.g., the window treatment assembly **110**) is in the extended position, for example, as shown in FIG. **2**. The extended position of the motorized window treatment **100** may include the first end **112** of the window treatment assembly **110** proximate to a first mounting bracket (e.g., mounting bracket **130**A) being further from a window and/or the structure to which the first mounting bracket is anchored than when the motorized window treatment **100** is in the operating position. The second end **114** (e.g., opposite the first end **112**) of the window treatment assembly **110** proximate to the second mounting bracket (e.g., mounting bracket **130**B) may remain substantially fixed when the motorized window treatment **100** is in the extended position, for example, as shown in FIG. **2**. Stated differently, the window treatment assembly **110** may pivot between the operating position and the extended position. The second end **114** of the window treatment assembly **110** and the mounting bracket **130**B may define a fulcrum about which the motorized window treatment **100** (e.g., the roller tube **111**) pivots.

When the motorized window treatment **100** is in the extended position, a motor drive unit housing end **150** may be exposed (e.g., accessible). The motor drive unit housing end **150** may be located proximate to the first end **112** of the window treatment assembly **110**. The motor drive unit housing end **150** may cover a cavity of the roller tube **111**. The motor drive unit housing end **150** may be configured to be removably secured to the roller tube **111** (e.g., at the first end **112** of the window treatment assembly **110**). For example, the motor drive unit housing end **150** may be configured to be secured within the cavity. The motor drive unit housing end **150** may be configured to retain one or more components (e.g., such as the batteries **260** shown in FIG. **3**).

The motor drive unit housing end **150** may include a control button **152**. The control button **152** may be backlit. For example, the control button **152** may include a light pipe (e.g., may be translucent or transparent) that is illuminated by a light emitting diode (LED) within the motor drive unit housing. The control button **152** may be configured to enable a user to change one or more settings of the motorized window treatment **100**. For example, the control button **152** may be configured to change one or more wireless communication settings and/or one or more drive settings. The control button **152** may be configured to enable a user to pair the motorized window treatment **100** with a remote control device to allow for wireless communication between the remote control device and a wireless communication circuit (e.g., an RF transceiver) of the motor drive unit **190**. The control button **152** may be configured to provide a status indication to a user. For example, the control button **152** may be configured to flash and/or change colors to provide the status indication to the user. The status indication may indicate when the motorized window treatment **100** is in a programming mode.

The motor drive unit housing end **150** may include a disable actuator **154** for detecting when the roller tube **111** is not in the operating position. The motor drive unit (e.g., the drive assembly) may be deactivated (e.g., automatically deactivated) when the roller tube **111** is not in the operating position. For example, the disable actuator **154** may be configured to disable the motor drive unit such that the covering material cannot be raised or lowered when the roller tube **111** is not in the operating position. The disable actuator **154** may disable a motor of the motor drive unit, for example, when the roller tube **111** is pivoted (e.g., or slid) from the operating position to the extended position. The disable actuator **154** may enable the motor when the roller tube **111** reaches the operating position. For example, the disable actuator **154** may be a button, a switch, and/or the like.

In addition, the motor drive unit housing end **150** may also comprise a position detect circuit (not shown) for detecting when the roller tube **111** is not in the operating position and deactivating (e.g., automatically deactivating) the drive assembly (e.g., rather than including the disable actuator **154**). For example, the position detect circuit may comprise a magnetic sensing circuit (e.g., a Hall-effect sensor circuit) configured to detect when the motor drive unit housing end **150** is in the extended position and not in close proximity to a magnet located inside of the mounting bracket **130**A. The position detect circuit may be configured to disable the drive assembly such that the covering material cannot be raised or lowered when the roller tube **111** is not in the operating position. The position detect circuit may disable a motor of the drive assembly, for example, when the roller tube **111** is pivoted (e.g., or slid) from the operating position to the extended position. The position detect circuit may enable the motor when the roller tube **111** reaches the operating position. For example, the position detect circuit may also comprise an IR sensor, a switch, and/or the like.

FIG. **3** depicts an example battery-powered motorized window treatment **200** (e.g., such as the motorized window treatment **100** shown in FIGS. **1** and **2**). The battery-powered motorized window treatment **200** may include a roller tube **210** (e.g., such the roller tube **111** shown in FIG. **1**), a flexible material **220** (e.g., a covering material) windingly attached to the roller tube **210**, a motor drive unit **290** (e.g., a drive assembly), and a plurality of batteries **260**. The battery-powered motorized window treatment **200** may further include a hembar **240** (e.g., such as the hembar **140** shown in FIGS. **1** and **2**) and one or more mounting brackets **230**A, **230**B (e.g., such as the mounting brackets **130**A, **130**B shown in FIGS. **1** and **2**). The motor drive unit **290** of the battery-powered motorized window treatment **200** may be powered by the batteries **260**. Although the battery-powered motorized window treatment **200** is shown with four batteries **260**, it should be appreciated that the battery-powered motorized window treatment **200** may include a greater or smaller number of batteries. The roller tube **210** may define a longitudinal axis **216**. The longitudinal axis **216** may extend along a longitudinal direction L.

The motor drive unit may comprise a housing **292** in which the batteries **260** may be housed. The housing **292** of the motor drive unit **290** may comprise a cap **250** that is configured to retain the batteries **260** within the housing **292** of the motor drive unit **290** (e.g., within the roller tube **210**). The cap **250** may define an outer surface **252** with a button **254** (e.g., such as button **152**. The button **254** may be backlit. For example, the button **254** may include a light pipe that is illuminated by an LED within the cap **250**. The button **254** may be configured to enable a user to change one or more settings of the battery-powered motorized window treatment **200** as similarly described with button **152**. The button **254** may be configured to enable a user to pair the battery-powered motorized window treatment **200** with a remote control device to allow for wireless communication between the remote control device and a wireless communication circuit of the motor drive unit **290**. The button **254** may be configured to provide a status indication to a user. For example, the button **254** may be configured to flash and/or change colors to provide the status indication to the user. The button **254** may indicate when the battery-powered motorized window treatment **200** is in a programming mode, for example, via the status indication.

The motor drive unit **290** may be at least partially received within the roller tube **210**. For example, the housing **292** of the motor drive unit **290** may define a battery compartment **211** (e.g., a cavity) that is configured to receive the batteries **260** of the motor drive unit **290**. The battery compartment **211** may be accessible when the battery-powered motorized window treatment **200** is in the extended position (e.g., pivoted) and the cap **250** is removed.

The motor drive unit **290** of the battery-powered motorized window treatment **200** may include a battery holder **270**. The battery holder **270** may be configured to keep the batteries **260** fixed in place securely while the batteries **260** are providing power to the motor drive unit **290**. The batteries **260** and the battery holder **270** may be configured to be removed from the battery compartment **211** of the housing **292** along the longitudinal axis **216** of the roller tube **210**. For example, the cap **250** may be removed (e.g., disengaged from the roller tube **210** and/or the housing **292** of the motor drive unit **290**) such that the batteries **260** and battery holder **270** can be accessed. The battery holder **270** may be configured to be translated (e.g., along the longitudinal axis **216** of the roller tube **210**) until it is removed from the housing **292**. The batteries **260** may remain within the battery holder **270** of the motor drive unit **290** when the battery holder **270** is removed from the battery compartment **211**. The batteries **260** may be removed from the battery holder **270** when it is removed from the battery compartment **211** of the housing **292**. Replacement batteries may be installed within the battery holder **270** while it is removed from the battery compartment **211** of the housing **292**. The battery holder **270** may be open at opposed ends, for example, such that the batteries **260** can be electrically connected to a printed circuit board of the motor drive unit **290**. For example, one of the batteries **260** (e.g., the battery distal from the end **213** of the roller tube **210** when the battery holder **270** is installed within the battery compartment **211** of the housing **292**) may be configured to abut a spring (e.g., such as spring **384** shown in FIG. **4**) within the housing **292** of the motor drive unit **290**. And, one of the batteries **260** (e.g., the battery proximate to the end **213** of the roller tube **210** when the battery holder **270** is installed within the battery compartment **211** of the housing **292**) may be configured to abut an electrical contact (e.g., the electrical contact **356** shown in FIG. **4**) within the cap **250**.

FIGS. **4** and **5** depict an example motorized window treatment **300** (e.g., such as the motorized window treatment **100** shown in FIGS. **1** and **2**, and/or the battery-powered motorized window treatment **200** shown in FIG. **3**). FIG. **4** is a front cross-sectional view and FIG. **5** is an enlarged front cross-sectional view of the motorized window treatment

300. The motorized window treatment **300** may include a window treatment assembly **310** having a roller tube **311** and a motor drive unit **390**. The roller tube **311** may be made from a conductive material, such as aluminum or other suitable metal. The motor drive unit **390** may be powered by one or more batteries **360**. Although not shown in FIGS. **4-5**, the window treatment assembly may also comprise a flexible material windingly attached to the roller tube **311** (e.g., such as the flexible material **120**, **220**) and an idler (e.g., such as the idler of the motorized window treatments **100**, **200**). The motorized window treatment **300** may also comprise one or more mounting brackets for mounting the motorized window treatment **300** to a structure, such as a first mounting bracket **330** (e.g., the first mounting bracket **130**A, **230**A) for supporting the motor drive unit **390** and a second mounting bracket (e.g., the second mounting bracket **130**B, **230**B) for supporting the idler. The first mounting bracket **330** and the second mounting bracket may be made from a conductive material, such as aluminum or other suitable metal. In addition, the first mounting bracket **330** and the second mounting bracket may be made from a non-conductive material, such as plastic.

The motorized window treatment **300** may be adjusted between an operating position (e.g., as shown in FIGS. **1**, **4**, and **5**) to an extended position (e.g., as shown in FIGS. **2** and **3**) while secured to the first mounting bracket **330** and the second mounting bracket. The operating position may be defined as a position in which the roller tube assembly **310** is supported by and aligned with the first mounting bracket **330** and the second mounting bracket (e.g., as shown in FIG. **1**). The motorized window treatment **300** may be configured to be operated between the operating position and an extended position, for example, to enable access to replace the batteries **360**. The extended position may be defined as a position in which a first end **312** of the roller tube assembly **310** is accessible while still attached to the first mounting bracket **330**. The extended position may define a pivoted position, for example, as shown in FIGS. **2** and **3**, where one of the first mounting bracket **330** extends such that the batteries **360** are accessible via the first end **312** of the roller tube assembly **310**.

The first mounting bracket **330** and the second mounting bracket may be configured to attach the motorized window treatment **300** to a structure. The first mounting bracket **330** may define a base (e.g., such as the base **132**) and an arm **334**. The base and the arm **334** may define a stationary portion of the first mounting bracket **330**. The first mounting bracket **330** may define a translating portion **336**. The translating portion **336** may include an attachment member **338** that is configured to receive an end of the window treatment assembly **310**. For example, the attachment member **338** of the mounting bracket **330** may be configured to receive the motor drive unit **390**. The attachment member **338** may define an aperture. The base may be configured to attach the first mounting bracket **330** to a structure. The structure may include a window frame (e.g., a head jamb or side jambs of a window frame), a wall, a ceiling, or other structure, such that the motorized window treatment **300** is mounted proximate to an opening (e.g., over the opening or in the opening), such as a window for example. When the first mounting bracket **330** is attached to a vertical structure, such as a wall, the arm **334** of the mounting bracket **330** may extend horizontally from the base.

The translating portion **336** may be configured to translate the window treatment assembly **310** between the operating position (e.g., as shown in FIG. **1**) and the extended position (e.g., as shown in FIG. **2**). The translating portion **336** may be proximate to the base when in the operating position and distal from the base when in the extended position. The first end **312** of the roller tube assembly **310** (e.g., an end of the motor drive unit **390**) may be accessible via the aperture (e.g., to replace the batteries **360**) when the translating portion **336** is in the extended position.

The arm **334** may define one or more features that enable the translating portion **336** to be translated between the operating position and the extended position while remaining attached thereto. The translating portion **336** may define one or more corresponding features that are configured to cooperate with the one or more features on the arm **334**. The arm **334** may define one or more slides **335** (e.g., an upper slide and a lower slide). The slides **335** may protrude from an inner surface of the arm **334**. The translating portion **336** may define one or more channels (e.g., an upper channel and a lower channel) that are configured to receive the slides **335**. The translating portion **336** may define a middle slide **339**, for example, between the channels. The arm **334** may define a channel (e.g., a middle channel) that is configured to receive the middle slide **339**. The slides **335**, **339** and the channels may define angled edges (e.g., tapered edges) such that the attachment of the translating position **336** to the arm **334** defines an interlocking slide, e.g., such as a dovetail slide. The translating portion **336** may translate along the slides **335** between the operating position and the extended position.

FIG. **6** is a perspective view of the motor drive unit **390** and FIG. **7** is an enlarged perspective view of an end portion of the motor drive unit **390**. The motor drive unit **390** may comprise a housing **380**. The housing **380** may comprise a body **381** and a cap **350** (e.g., the cap **250**). The body **381** may be cylindrical. The cap **350** may be configured to attach to the body **381**. The housing **380** may comprise an upper portion **382**A and a lower portion **382**B. The motor drive unit **390** may be operatively coupled to the roller tube **311**, for example, via a coupler **395** (e.g., a drive coupler). The coupler **395** may be an output gear that is driven by a motor **396** and transfers rotation of the motor **396** to the roller tube **311**. For example, the coupler **395** may define a plurality of splines **397** about its periphery. An inner surface of the roller tube **311** may be grooved. That is, the inner surface of the roller tube **311** may define a plurality of grooves (not shown). The splines **397** of the coupler **395** may be configured to engage respective groove in the roller tube **311** such that rotation of the motor **396** is transferred to the roller tube **310**, for example, via the coupler **395**.

The motorized window treatment **300** (e.g., the motor drive unit **390**) may include a bearing assembly **320** having an inner bearing **322** and an outer bearing **324** that are located external to the roller tube **311**. The inner bearing **322** and the outer bearing **324** may be non-metallic (e.g., plastic) sleeve bearings. The bearing assembly **320** may be captured between the roller tube **311** and the mounting bracket **330**, such that the bearing assembly **320** may be located in a gap **328** (e.g., longitudinal gap) between the roller tube **311** and the mounting bracket **330**. The components of the motorized window treatment **300** in and/or adjacent to the gap **328** may be non-conductive such that radio-frequency field disruption and/or shielding is minimized. For example, the motorized window treatment **300** may not include conductive (e.g., metal) components in an area radially surrounding the gap **328**. The inner bearing **322** may engage the housing **380** of the motor drive unit **390**. The inner bearing **322** may be operatively coupled to the motor drive unit housing **380**. FIG. **8** is an enlarged perspective view of the end portion of the motor drive unit **390** with the bearing assembly **320** removed. For example, the inner bearing **322** may define splines (not shown) that are configured to be received by grooves **386** around the periphery of the housing **380** of the motor drive unit **390**. The inner bearing **322** may be press fit onto the housing **380** of the motor drive unit **390**. The outer bearing **324** may engage the roller tube **311**. The outer bearing **324** may be operatively coupled to the roller tube **311**. The outer bearing **324** may rotate with the roller tube **311**. The outer bearing **324** may be press fit into engagement with the roller tube **311**. For example, the outer bearing **324** may comprise a plurality of splines **326** that are configured to engage grooves (not shown) of the roller tube **311**. The inner bearing **322** may remain stationary with the motor drive unit housing **380** as the roller tube **311** rotates. Stated differently, the roller tube **311** and the outer bearing **324** may rotate about the inner bearing **322** and the housing **380** of the motor drive unit **390**.

The motor drive unit **390** may include a battery holder **370** (e.g., the battery holder **270**). The battery holder **370** and the cap **350** may keep the batteries **360** fixed in place securely while the batteries **360** are providing power to the motor drive unit **390** and/or the cap **350**. The battery holder **370** may be configured to clamp the batteries **360** together (e.g., as shown in FIG. **3**) such that the batteries **360** can be removed from the motorized window treatment **300** at the same time (e.g., together). The battery holder **370** may be received in a motor drive unit cavity **389** of the motor drive unit **390**. The motor drive unit **390** may be received within a roller tube cavity **315** (FIG. **4**). The roller tube cavity **315** may be open proximate to an end **313** of the roller tube **311** (e.g., the first end **312** of the window treatment assembly **310**). The batteries **360** may be configured to be removed from the motor drive unit **390**, for example, while the housing **380** of the motor drive unit **390** remains engaged with the first mounting bracket **330** and the second mounting bracket. That is, the batteries **360** may be configured to be removed from the motor drive unit **390** when the motorized window treatment **300** is in the pivoted position. An inside diameter of the inner bearing **322** may be greater than an outer diameter of the batteries **360** and/or the battery holder **370**.

The cap **350** may be configured to cover an end of the motor drive unit cavity **389**. For example, the cap **350** may be received (e.g., at least partially) within the motor drive unit cavity **389**. The cap **350** may include a button **352**, a control interface printed circuit board **354**, and an electrical contact **356** (e.g., a conductive pad) electrically coupled to the control interface printed circuit board **354**. The electrical contact **356** may be a positive electrical contact, for example, as shown in FIG. **5**. Alternatively, the electrical contact **356** may be a negative electrical contact. The cap **350** may include a switch **355** (e.g., a mechanical tactile switch) mounted to the control interface printed circuit board **354** and configured to be actuated in response to actuations of the button **352**. The button **352** may be illuminated by a light-emitting diode (LED) **358** mounted to the control interface printed circuit board **354**.

The motor drive unit **390** may include a motor drive printed circuit board **392**, an intermediate storage device **394**, and a gear assembly **398**. The intermediate storage device **394** may include one or more capacitors (e.g., super capacitors) and/or one or more rechargeable batteries. The batteries **360** may be located between the cap **350** and the motor drive printed circuit board **392** of the motor drive unit **390**. The motor drive unit **390** may also comprise a motor drive circuit (e.g., such as the motor drive circuit **820** shown in FIG. **16**) for driving the motor **396**, a control circuit (e.g., such as the control circuit **830** shown in FIG. **16**) for controlling the motor drive circuit, and/or, a wireless communication circuit **399** (e.g., such as the communication circuit **842** shown in FIG. **16**) mounted to the motor drive printed circuit board **392**.

The motor drive unit **390** may include a spring **384**, which may extend from an internal wall of the motor drive unit cavity **389**. The spring **384** may be configured to abut and apply a force to one of the batteries **360**, for example, such that the batteries **360** remain in contact with one another while installed within the motor drive unit cavity **389**. The spring **384** may be electrically coupled to the motor drive printed circuit board **392** via a wire **385**. The spring **384** may be a negative electrical contact, for example, as shown in FIG. **4**. Alternatively, the spring **384** may be a positive electrical contact. The spring **384** may be configured to apply a force to the batteries **360** to maintain electrical connection of the batteries **360** with the spring **384** and the electrical contact **356** of the cap **350**.

The button **352** may be configured to enable a user to change one or more settings of the motorized window treatment **300**. For example, the button **352** may be configured to change one or more settings of the control interface printed circuit board **354** and/or the motor drive printed circuit board **392**. The button **352** may be configured to enable a user to pair the motorized window treatment **300** with a remote control device to allow for wireless communication between the remote control device and the wireless communication circuit **399** mounted to the motor drive printed circuit board **392**. The button **352** may be configured to provide a status indication to a user. For example, the control button **352** may be configured to flash and/or change colors to provide the status indication to the user. The button **352** may be configured to indicate (e.g., via the status indication) whether the motor drive unit **390** is in a programming mode.

The control interface printed circuit board **354** and the motor drive printed circuit board **392** may be electrically connected. For example, the motorized window treatment **300** may include a ribbon cable **386**. The ribbon cable **386** may be attached to a connector **388** mounted to the control interface printed circuit board **354** and a similar connector mounted to the motor drive printed circuit board **392**. The ribbon cable **386** may be configured to electrically connect the control interface printed circuit board **354** and the motor drive printed circuit board **392**. The ribbon cable **386** may terminate at the control interface printed circuit board **354** and the motor printed circuit board **392**. For example, the ribbon cable **386** may extend within the motor drive unit cavity **389**. The ribbon cable **386** may include electrical conductors for providing power from the batteries **360** to the control interface printed circuit board **354** and/or the motor drive printed circuit board **392**. The ribbon cable **386** may include electrical conductors for conducting control signals (e.g., for transmitting one or more messages) between the control interface printed circuit board **354** and the motor drive printed circuit board **392**. For example, the ribbon cable **386** may be configured to conduct power and/or control signals between the control interface printed circuit board **354** and the motor drive printed circuit board **392**.

The motor drive unit **390** may further comprise an antenna **400** (e.g., as shown in FIG. **5**). For example, the antenna **400** may comprise an insulated electrical conductor, such as a 22-gauge stranded electrical wire with a polyvinyl chloride (PVC) coating. The antenna **400** may be wrapped around (e.g., wound about) the body **381** of the housing **380**. The antenna **400** may be located in one or more channels, such as first and second peripheral (e.g., circumferential) channels **411**, **412**, in the housing **380** of the motor drive unit **390**. The first peripheral channel **411** may comprise a first portion **411**A formed in an outer surface **410**A of the upper portion **382**A of the housing **380** and a second portion **411**B formed in an outer surface **410**B of the lower portion **382**B of the housing **380**. The second peripheral channel **412** may comprise a first portion **412**A formed in the outer surface **410**A of the upper portion **382**A of the housing **380** and a second portion **412**B formed in the outer surface **410**B of the lower portion **382**B of the housing **380**. The first and second peripheral channels **411**, **412** may extend around the circumference of the housing **380** of the motor drive unit **390**, such that the antenna **400** may be wrapped around the motor drive unit **390**.

As shown in FIG. **5**, the antenna **400** may be wrapped around the housing **380** of the motor drive unit **390** adjacent to the end **313** of the roller tube **311**. For example, the antenna **400** may be wrapped around the housing **380** of the motor drive unit **390** below the bearing assembly **320**, e.g., such that the antenna **400** is located within an area that surrounds the circumference of the motor drive unit and falls within an area defined by the bearing assembly **320**. The antenna **400** may be wrapped around the motor drive unit **390** adjacent to the gap **328** between the roller tube **311** and the mounting bracket **330** (e.g., lie at least partially within the area defined by the gap). For example, the antenna **400** may be aligned with the gap **328**. The antenna **400** may transmit and/or receive RF signals through the gap **328**. In addition, when the antenna **400** is emitting electromagnetic waves, the electromagnetic waves may be coupled to the roller tube **311** (e.g., capacitively coupled to the roller tube), which may result in current flow (e.g., standing waves) on the surface of the roller tube **611** (e.g., since the roller tube **311** is made from a conductive material). As a result, the roller tube **311** may re-radiate the electromagnetic waves emitted by the antenna **400**, which may increase the amount of RF signals transmitted and/or received by the antenna **400**.

A distance (e.g., in the longitudinal direction L) between windings of the antenna **400** may be configured to prevent the antenna **400** from coupling to itself. For example, the distance between adjacent windings of the antenna **400** in the first peripheral channel **411** and the second peripheral channel **412** may be approximately 0.1 to 0.4 inches (e.g., such as 0.2 inches). The first peripheral channel **411** may be located closer to the roller tube **311** than the second peripheral channel **412**. For example, the first peripheral channel **411** may be close to an outer edge of the roller tube **311** and partially underneath of the roller tube **311** (e.g., but not fully underneath of the roller tube **311**) as shown in FIG. **5**. For example, the second peripheral channel **412** may be located as far away from the first peripheral channel **411** as possible, but not underneath or within the attachment portion **338** of the mounting bracket **330**. A center of the first peripheral channel **411** may be, for example, approximately 0.20 inches away from a center of the second peripheral channel **412**. The first and second portions **382**A, **382**B of the housing **380** may be identical, such that two separate types of unique housing parts are not required to form the housing **380** of the motor drive unit **390**. Since only one type of housing portion is required to form the housing **380**, a manufacturer of the motor drive unit **390** is able to stock less parts in inventory.

FIG. **9** is a partial exploded view of the motor drive unit **390** looking up into the inside of the upper portion **382**A of the housing **380**. The motor drive unit **390** may comprise a coupling printed circuit board **420** for coupling the antenna **400** to the wireless communication circuit **399** mounted to the motor drive printed circuit board **392**. The coupling printed circuit board **420** may be received in a recess **422** in an inner surface **424** of the upper portion **382**A of the housing **380**. For example, the coupling printed circuit board **420** may be held in the recess **422** by one or more snaps **425**. The coupling printing circuit board **420** may have a matching network circuit **426** mounted thereto. The matching network circuit **426** may be electrically coupled between the antenna **400** and the wireless communication circuit **399** mounted to the motor drive printed circuit board **392**. The matching network circuit **426** may be electrically coupled to the wireless communication circuit **399** on the motor drive printed circuit board **392** via a coaxial cable **430**, which may be electrically coupled to the coupling printed circuit board **420** via a coaxial connector **428**. The matching network circuit **426** may be configured to optimize the performance of the antenna **400**. For example, the matching network circuit **426** may be configured to match an impedance of the antenna **400** to an impedance of the wireless communication circuit **399** to obtain a maximum transfer of power between the antenna **400** and the wireless communication circuit **399**. The matching network circuit **426** may include, for example, an inductor-capacitor (LC) filter. The coaxial cable **430** may extend through a coaxial cable channel **432** formed in the inner surface **424** of the upper portion **382**A of the housing **380** and may be held in place by one or more tabs **434**. The coaxial cable channel **432** may extend in the longitudinal direction L of the motorized window treatment **300**. In addition, the wireless communication circuit **399** may be mounted to the coupling printed circuit board **420** and the wireless communication circuit **399** may be coupled to the circuitry on the motor drive printed circuit board **392** via a cable, such as a ribbon cable (e.g., like the ribbon cable **386**).

As shown in FIG. **5**, the antenna **400** may terminate at the coupling printed circuit board **430**. For example, the antenna **400** may be received through a through-hole **439** in the coupling printed circuit board **430** and be soldered to an electrical pad surrounding the through-hole **439** to electrically couple the antenna **400** to the matching network circuit **436**. The antenna **400** may extend from the first peripheral channel **411** (e.g., the first portion **411**A in the outer surface **411**A of the upper portion **382**A of the housing **380**) through an intermediate channel **415** and an opening **416** to the coupling printed circuit board **430**. The intermediate channel **415** may also be formed in the outer surface **410**A of the upper portion **382**A of the housing **380** and may extend in the longitudinal direction L of the motorized window treatment **300**. The first and second peripheral channels **411**, **412** and the intermediate channel **415** may meet (e.g., intersect) at a recess **414**, which may also be formed in the outer surface **411**A of the upper portion **382**A of the housing **380**.

As the antenna **400** exits the intermediate channel **415** from coupling printed circuit board **430** and enters the recess **414**, the electrical wire of the antenna **400** may follow a first path **418**A to extend along the first peripheral channel **411**. A corner **417** adjacent to the antenna **400** in the recess **414** may be rounded to facilitate bending of the electrical wire of the antenna **400**. The antenna **400** may extend through the first peripheral channel **411** and fully wrap around the circumference of the housing **380** and enter the recess **414** again. After exiting the first peripheral channel **411** and entering the recess **414**, the electrical wire of the antenna **400** may extend along a second path **418**B (e.g., diagonally) across the recess **414** and enter the second peripheral channel **412**. The antenna **400** may extend through the second peripheral channel **412** and wrap (e.g., partially or fully wrap) around the housing **380** and terminate in the second periphery channel **412**. The electrical wire of the antenna **400** may be held in the first and second peripheral channels **411**, **412** and the intermediate channel **415** by tabs **419**.

When the wireless communication circuit **399** drives the antenna **400** with a signal, the antenna **400** may emit electromagnetic waves (e.g., radio-frequency signals). As previously mentioned, the antenna **400** may be wrapped around the motor drive unit **390** underneath of the bearing assembly **320** (e.g., which is made from a non-conductive material, such as plastic) and adjacent to the gap **328** between the roller tube **311** and the mounting bracket **330**. When the antenna **400** is emitting electromagnetic waves, the electromagnetic waves may be coupled to the roller tube **311** (e.g., which may be made of a conductive material, such as aluminum), which may result in current flow on the surface of the roller tube **311**. As a result, the roller tube **311** may re-radiate the electromagnetic waves emitted by the antenna **400**.

While the antenna **400** is shown in FIGS. **4-9** being wrapped around the motor drive units **390** one or more times, the antenna **400** may be wrapped around the motor drive unit **390** in other manners and/or shapes. For example, the one or more channels in which the antenna **400** is located may be formed on the inner surface **424** of the upper portion **382**A of the housing **380**. In addition, the antenna **400** may be internal to the housing **380** of the motor drive unit **390** (e.g., extending through one or more tunnels in the housing **380**). Alternatively, the one or more channels in which the antenna **400** is located may be formed in the bearing assembly **320** (e.g., such as the stationary bearing **322**).

FIG. **10** is an enlarged front cross-sectional view of another example motorized window treatment **300'** (e.g., such as the motorized window treatment **100** shown in FIGS. **1** and **2**, the battery-powered motorized window treatment **200** shown in FIG. **3**, and/or the motorized window treatment **300** shown in FIGS. **4** and **5**). The motorized window treatment **300'** may include the window treatment assembly **310** having the roller tube **311** and the motor drive unit **390**. The motor drive unit **390** may be powered by the one or more batteries **360**. Although not shown in FIG. **10**, the window treatment assembly **310** may also comprise a flexible material windingly attached to the roller tube **311** (e.g., such as the flexible material **120**, **220**) and an idler (e.g., such as the idler of the motorized window treatments **100**, **200**). The motorized window treatment **300'** may also comprise one or more mounting brackets for mounting the motorized window treatment **300** to a structure, such as the first mounting bracket **330** (e.g., the first mounting bracket **130**A, **230**A) for supporting the motor drive unit **390** and the second mounting bracket (e.g., the second mounting bracket **130**B, **230**B) for supporting the idler.

FIG. **11** is a top view of a flexible printed circuit board **440'** of the motor drive unit **390** of the motorized window treatment **300'** (e.g., shown in an unwound state). The motor drive unit **390** may comprise an antenna **400'** that may be formed on the flexible printed circuit board **440'**. For example, the flexible printed circuit board **440'** may replace both the coupling printed circuit board **420** and the electrical wire of the antenna **400** (e.g., of the motorized window treatment **300** shown in FIGS. **4** and **5**). The flexible printed circuit board **440'** may comprise an antenna portion **421'** (e.g., an elongated portion) and a coupling portion **420'**. The antenna **400'** may comprise an electrical trace (e.g., an electrical conductor) that extends through the antenna portion **421'** of the flexible printed circuit board **440'** and may operate as the antenna to radiate the RF signals. The antenna **400'** (e.g., the antenna portion **421'** of the flexible printed circuit board **440'**) may be located in one or more channels, such as first and second peripheral (e.g., circumferential) channels **411**, **412**, in the housing **380** of the motor drive unit **390**. The first peripheral channel **411** may comprise a first portion **411**A formed in an outer surface **410**A of the upper portion **382**A of the housing **380** and a second portion **411**B formed in an outer surface **410**B of the lower portion **382**B of the housing **380**. The second peripheral channel **412** may comprise a first portion **412**A formed in the outer surface **410**A of the upper portion **382**A of the housing **380** and a second portion **412**B formed in the outer surface **410**B of the lower portion **382**B of the housing **380**. The first and second peripheral channels **411**, **412** may extend around the circumference of the housing **380** of the motor drive unit **390**, such that the antenna **400'** (e.g., the antenna portion **421'** of the flexible printed circuit board **440'**) may be wrapped around the motor drive unit **390**.

As shown in FIG. **11**, the antenna portion **421'** of the flexible printed circuit board **440'** may extend from the coupling portion **420'**. The antenna portion **421'** of the flexible printed circuit board **440'** may define first and second portion **402'**, **404'** that are configured to be wound about the housing **380** of the motorized window treatment **300'**. The antenna portion **421'** of the flexible printed circuit board **440'** may define a third portion **406'** that connects the first and second portions **402'**, **404'**. The third portion **406'** may be substantially perpendicular to the first and second portions **402'**, **404'**, for example, such that the first and second portions **402'**, **404'** are spaced apart (e.g., by a distance in the longitudinal direction L defined by a length of the third portion **406'**) when the antenna **400'** is wound about the housing **380** of the motorized window treatment **300'**. For example, the third portion **406'** may extend through the recess **414** between the first and second peripheral channels **411**, **412** when the antenna **400'** is wound about the housing **380** of the motorized window treatment **300'**.

The matching network circuit **426** may be mounted to the coupling portion **420'** of the flexible printed circuit board **420'**. For example, the coupling portion **420'** of the flexible printed circuit board **440'** may be located in the recess **422** in the inner surface **424** of the upper portion **382**A of the housing **380**. The antenna portion **421'** of the flexible printed circuit board **440'** may also define an intermediate portion **408'** that connects the first portion **402'** of the antenna portion **421'** to the coupling portion **420'**. For example, the intermediate portion **408'** may extend through the intermediate channel **415** between the recess **422** and the first peripheral channel **411** when the antenna **400'** is wound about the housing **380** of the motorized window treatment **300'**. The antenna portion **421'** may extend from the matching network circuit **426** on the coupling portion **420'** of the flexible printed circuit board **440'** in the recess **422** through the opening **415**, the intermediate channel **415**, and the first and second peripheral channels **411**, **412**. Since the flexible printed circuit board **420'** is flexible, the antenna portion **421'** may be configured to wrap around the housing **380** as the antenna portion **421'** extends through the first and second peripheral channels **411**. When the antenna portion **421'** is wrapped around (e.g., wound about) the housing **380**, the third portion **406'** and the intermediate portion **408'** may extend in the longitudinal direction L. The matching network circuit **426** may be electrically coupled to the motor drive printed circuit board **392** via the coaxial cable **430** that may be connected to the coaxial connector **428** and may extend through the coaxial cable channel **432**. The antenna **400'** may be electrically coupled to the wireless communication circuit **399** on the motor drive printed circuit board **392** via the matching network circuit **426** on the coupling portion **420'** of the flexible printed circuit board **440'**, such that the electrical trace on the antenna portion **421'** of the flexible printed circuit board **440'** may radiate the RF signals.

As shown in FIG. **10**, the antenna **400'** (e.g., the antenna portion **421'** of the flexible printed circuit board **440'**) may be wrapped around the housing **380** of the motor drive unit **390** adjacent to the end **313** of the roller tube **311**. For example, the antenna **400'** may be wrapped around the housing **380** of the motor drive unit **390** below the bearing assembly **320**, e.g., such that the antenna **400'** is located within an area that surrounds the circumference of the motor drive unit and falls within an area defined by the bearing assembly **320**. The antenna **400'** may be wrapped around the motor drive unit **390** adjacent to the gap **328** between the roller tube **311** and the mounting bracket **330** (e.g., lie at least partially within the area defined by the gap). For example, the antenna **400'** may be aligned with the gap **328**. The antenna **400'** may transmit and/or receive RF signals through the gap **328**. In addition, when the antenna **400'** is emitting electromagnetic waves, the electromagnetic waves may be coupled to the roller tube **311** (e.g., capacitively coupled to the roller tube), which may result in current flow (e.g., standing waves) on the surface of the roller tube **311** (e.g., since the roller tube **311** is made from a conductive material). As a result, the roller tube **311** may re-radiate the electromagnetic waves emitted by the antenna **400'**, which may increase the amount of RF signals transmitted and/or received by the antenna **400'**.

A distance (e.g., in the longitudinal direction L) between windings of the antenna **400'** may be configured to prevent the antenna **400'** from coupling to itself. For example, the first portion **402'** of the antenna **400'** may be spaced apart by the distance from the second portion **404'** of the antenna **400'**, when the antenna **400'** (e.g., the electrical trace of the antenna portion **421'** of the flexible printed circuit board **440'**) is received within the first and second peripheral channels **411**, **412**. For example, the distance between adjacent windings of the antenna **400'** in the first peripheral channel **411** and the second peripheral channel **412** may be approximately 0.1 to 0.4 inches (e.g., such as 0.2 inches). The first peripheral channel **411** may be located closer to the roller tube **311** than the second peripheral channel **412**. For example, the first peripheral channel **411** may be close to an outer edge of the roller tube **311** and partially underneath of the roller tube **311** (e.g., but not fully underneath of the roller tube **311**) as shown in FIG. **10**. For example, the second peripheral channel **412** may be located as far away from the first peripheral channel **411** as possible, but not underneath or within the attachment portion **338** of the mounting bracket **330**. A center of the first peripheral channel **411** may be, for example, approximately 0.20 inches away from a center of the second peripheral channel **412**.

FIG. **12** is an enlarged front cross-sectional view of another example motorized window treatment **300"**. The motorized window treatment **300"** may comprise a motor drive unit **390'** having an antenna **500**. The motorized window treatment **300"** and the motor drive unit **390'** may have many similar components as the motorized window treatment **300** and the motor drive unit **390** shown in FIGS. **2-9**, respectively. The motor drive unit **390'** may comprise a housing **380'**. The housing **380'** may comprise a body **381'** and a cap **350** (e.g., the cap **250**). The body **381'** may be cylindrical. The cap **350** may be configured to attach to the body **381**. The body **381'** may comprise an upper portion **382**A' and a lower portion **382**B'. FIG. **13** is an enlarged perspective view of an end portion of the motor drive unit **390'** with a bearing assembly (e.g., the bearing assembly **320**) removed. Rather than comprising the first and second peripheral channels **411**, **412** that are parallel to each other and extend around the circumference of the motor drive unit **390**, the motor drive unit **390'** may comprise a spiral-shaped channel **511** that extends around the housing **380'** (e.g., the body **381'**) of the motor drive unit **390'** (e.g., about the periphery of the housing **380'**). The spiral-shaped channel **511** may be a continuous spiral that moves farther and farther away, in the longitudinal direction L, from the roller tube **311**. The antenna **500** may be located within the spiral-shaped channel **511**, such that the antenna **500** may be wrapped around the motor drive unit **390'**. For example, the antenna **500** may be received within the spiral-shaped channel **511**. A distance (e.g., in the longitudinal direction L) between windings of the antenna **500** may be configured to prevent the antenna **500** from coupling to itself. For example, the distance between adjacent windings (e.g., wound portions) of the antenna **500** in the spiral-shaped channel **511** may be approximately 0.1 to 0.4 inches (e.g., such as 0.2 inches). The antenna **500** may comprise an insulated electrical conductor, for example, such as a 22-gauge stranded electrical wire with a polyvinyl chloride (PVC) coating.

The upper portion **382**A' of the housing **382'** may comprise a first channel portion **511**A, a second channel portion **512**A, a third channel portion **513**A, and a fourth channel portion **514**A formed in an outer surface **510**A of the upper portion **382**A'. The lower portion **382**B' of the housing **382'** may be identical to the upper portion **382**A' and may comprise a first channel portion **511**B, a second channel portion **512**B, a third channel portion **513**B, and a fourth channel portion (not shown) formed in an outer surface **510**B of the lower portion **382**B'. The fourth channel portion of the lower portion **382**B' of the housing **382'** may be similar (e.g., identical) to the fourth channel portion **514**A of the upper portion **382**A' and may be aligned with the third channel portion **513**A of the upper portion **382**A'. The spiral-shaped channel **511** in which the antenna **500** resides may include the first channel portion **511**A of the upper portion **382**A', the second channel portion **512**B of the lower portion **382**B', the third channel portion **513**A of the upper portion **382**A', and the fourth channel portion of the lower portion **382**B'.

The upper portion **382**A' of the housing **382'** may comprise a second spiral-shaped channel **512** which remains vacant when the antenna **500** is spirally wound about the motor drive unit **390'** may include the second channel portion **512**A of the upper portion **382**A', the first channel portion **511**B of the lower portion **382**B', and the third channel portion **513**B of the lower portion **382**B', and the fourth channel portion **514**A of the upper portion **382**A'. The first and second portions **382**A', **382**B' of the housing **380'** may be identical, such that two separate types of unique housing parts are not required to form the housing **380'** of the motor drive unit **390'**. Since only one type of housing portion is required to form the housing **380'**, a manufacturer of the motor drive unit **390'** is able to stock less parts in inventory. Because the upper portion **382**A' and the lower portion **382**B' are identical, the housing **380'** may comprise a second spiral-shaped channel formed by the first channel portion **511**B of the lower portion **382**B', the second channel portion **512**A of the upper portion **382**A', the third channel portion **513**B of the lower portion **382**B', and the fourth channel portion **514**A of the upper portion **382**A'.

Although the body **380'** of the housing **380'** of the motor drive unit **390'** is shown in the figures as having an upper portion **382**A' and a lower portion **382**B' that are identical, it should be appreciated that the body **381'** could alternatively be a single piece, two non-identical pieces, or more than two pieces. While the antenna **500** is shown in FIGS. **12** and **13** being wrapped around the housing **380'** (e.g., the body **380'**) of the motor drive unit **390'** one or more times, the antenna **500** may be wrapped around the motor drive unit **390'** in other manners and/or shapes. For example, the spiral-shaped channel **511** in which the antenna **500** is located may be formed on the inner surface **424** of the upper portion **382**A' of the body **380"**. In addition, the antenna **500** may be internal to the housing **380'** (e.g., the body **380'**) of the motor drive unit **390'** (e.g., extending through one or more tunnels in the housing **380'**). Although the spiral-shaped channel **511** is shown on the outer surfaces **510**A, **510**B of the housing **380'**, it should be appreciated that the spiral-shaped channel **511** could instead be located on the bearing assembly **320** (e.g., such as the stationary bearing **322**).

As shown in FIG. **12**, the antenna **500** may be wrapped around the motor drive unit **390'** below the bearing assembly **320**, e.g., such that the antenna **400** is located within an area that surrounds the circumference of the motor drive unit **390'** and falls within an area defined by the bearing assembly **320**. The antenna **500** may be wrapped around (e.g., spirally wound about) the motor drive unit **390'** adjacent to the gap **328** between the roller tube **311** and the mounting bracket **330** (e.g., located at least partially within the area defined by the gap **328**). For example, the antenna **500** may be aligned with the gap **328**. The antenna **500** may transmit and/or receive RF signals through the gap **328**. Because the components (e.g., all of the components) of the motorized window treatment **300** in and/or adjacent to the gap **328** are non-conductive, interference with and/or disruption of the RF signals being transmitted and/or received by the antenna **500** may be minimized (e.g., prevented). The spiral-shaped channel **511** may continuously move the antenna **500** away from the roller tube **311** in the longitudinal direction L as the antenna **500** follows the spiral-shaped channel **511**.

FIG. **14** is a partially exploded view of the motor drive unit **390'** looking up into the inside of the upper portion **382**A' of the housing **380'**. The motor drive unit **390'** may comprise a coupling printed circuit board **520** for coupling the antenna **500** to the wireless communication circuit **399** mounted to the motor drive printed circuit board **392**. The coupling printed circuit board **520** may be received in a recess **522** in an inner surface **524** of the upper portion **382**A' of the housing **380'**. For example, the coupling printed circuit board **520** may be held in the recess **522** by one or more snaps **525**. The coupling printing circuit board **520** may have a matching network circuit **526** mounted thereto. The matching network circuit **526** may be electrically coupled between the antenna **500** and the wireless communication circuit **399** mounted to the motor drive printed circuit board **392**. The matching network circuit **526** may be electrically coupled to the wireless communication circuit **399** on the motor drive printed circuit board **392** via a coaxial cable **530**, which may be electrically coupled to the coupling printed circuit board **520** via a coaxial connector **528**. The matching network circuit **526** may be configured to optimize the performance of the antenna **500**. For example, the matching network circuit **526** may be configured to match an impedance of the antenna **500** to an impedance of the wireless communication circuit **399** to obtain a maximum transfer of power between the antenna **500** and the wireless communication circuit **399**. The matching network circuit **526** may include, for example, an inductor-capacitor (LC) filter. The coaxial cable **530** may extend through a coaxial cable channel **532** formed in the inner surface **524** of the upper portion **382**A' of the housing **380'**. The coaxial cable **530** may be held in place within the coaxial cable channel **532** by one or more tabs **534**. The coaxial cable channel **532** may extend in the longitudinal direction L of the motorized window treatment **300'**. In addition, the wireless communication circuit **399** may be mounted to the coupling printed circuit board **520** and the wireless communication circuit **399** may be coupled to the circuitry on the motor drive printed circuit board **392** via a cable, such as a ribbon cable (e.g., like the ribbon cable **386**).

As shown in FIG. **12**, the antenna **500** may terminate at the coupling printed circuit board **530**. For example, the antenna **500** may be received through a through-hole **539** in the coupling printed circuit board **530**. The antenna may be soldered to an electrical pad surrounding the through-hole **539**, for example, to electrically couple the antenna **500** to the matching network circuit **536**. The antenna **500** may extend from the spiral-shaped channel **511** (e.g., the first portion **511**A in the upper portion **382**A' of the housing **380'**) through an intermediate channel **515** and an opening **516** to the coupling printed circuit board **530**. The intermediate channel **515** may also be formed in the outer surface **510**A of the upper portion **382**A' of the housing **380** and may extend in the longitudinal direction L of the motorized window treatment **300'**.

As the antenna **500** exits the intermediate channel **515** from coupling printed circuit board **530** and enters the spiral-shaped channel **511**, the electrical wire of the antenna **500** may follow a first path **518**A to extend along the first channel portion **511**A in the upper portion **382**A'. A corner **517** adjacent to the antenna **500** in the first channel portion **511**A of the upper portion **382**A' may be rounded to facilitate bending of the electrical wire of the antenna **500**. The antenna **500** may extend through the spiral-shaped channel **511** and fully wrap around the circumference of the housing **380'** one or more times. The antenna **500** may extend through the first channel portion **511**A of the upper portion **382**A', the second channel portion **512**B of the lower portion **382**B', the third channel portion **513**A of the upper portion **382**A', and the fourth channel portion of the lower portion **382**B'. For example, the antenna **500** may extend along a second path **518**B from the second channel portion **512**B of the lower portion **382**B' to the third channel portion **513**A of the upper portion **382**A'. Stated differently, the second path **518**B of the antenna **500** may extend from the second channel portion **512**B of the lower portion **382**B' to the third channel portion **513**A of the upper portion **382**A'. The antenna **500** may or may not extend for the full length of the spiral-shaped channel **511**. For example, the antenna **500** may not extend into the fourth channel portion of the lower portion **382**B'. The electrical wire of the antenna **500** may be held in the spiral-shaped channel **511** and the intermediate channel **515** by tabs **519**. For example, the tabs **519** may be configured to retain the antenna **500** within the spiral-shaped channel **511**.

When the wireless communication circuit **399** drives the antenna **500** with a signal, the antenna **500** may emit electromagnetic waves (e.g., radio-frequency signals). As previously mentioned, the antenna **500** may be wrapped around the motor drive unit **390'** underneath of the bearing assembly **320** (e.g., which is made from a non-conductive material, such as plastic) and adjacent to the gap **328** between the roller tube **311** and the mounting bracket **330**.

When the antenna **500** is emitting electromagnetic waves, the electromagnetic waves may be coupled to the roller tube **311** (e.g., which may be made of a conductive material, such as aluminum), which may result in current flow on the surface of the roller tube **311**. As a result, the roller tube **311** may re-radiate the electromagnetic waves emitted by the antenna **500**. Because the antenna **500** extends through the spiral-shaped channel **511**, the antenna **500** may more quickly move in distance away from the roller tube, which may allow more electromagnetic waves to couple to the roller tube **311** (e.g., as compared to extending through the first and second peripheral channels **411**, **412**).

In some examples, the antenna **500** may be formed on a flexible printed circuit board (not shown). The flexible printed circuit board may replace both the coupling printed circuit board **520** and the electrical wire of the antenna. The matching network circuit **526** may be mounted to the flexible printed circuit board, for example, to a coupling portion of the flexible printed circuit board that is located in the recess **522** in the inner surface **524** of the upper portion **382**A' of the housing **380'**. The flexible printed circuit board may comprise an antenna portion (e.g., an elongated portion) having an electrical trace (e.g., an electrical conductor) that may operate as the antenna to radiate the RF signals. The antenna portion may extend from the matching network circuit **526** on the portion of the flexible printed circuit board in the recess **522** through the opening **515**, the intermediate channel **515**, and the spiral-shaped channel **511**. Since the flexible printed circuit board is flexible, the antenna portion may be configured to wrap around the housing **380'** as the antenna portion extends through the first and second peripheral channels **511**. The matching network circuit **526** may be electrically coupled to the motor drive printed circuit board **392** via the coaxial cable **530** that extends through the coaxial cable channel **532**.

FIG. **15** is an enlarged front cross-sectional view of another example motorized window treatment **600**. The motorized window treatment **600** may include a window treatment assembly (e.g., the window treatment assembly **310**) having a roller tube **611** (e.g., the roller tube **311**) and a motor drive unit **690**. The roller tube **611** may be made from a conductive material, such as aluminum or other suitable metal. The motor drive unit **690** may be powered by one or more batteries **660** (e.g., the batteries **360**). Although not shown in FIGS. **4-5**, the window treatment assembly may also comprise a flexible material windingly attached to the roller tube **611** (e.g., such as the flexible material **120**, **220**) and an idler (e.g., such as the idler of the motorized window treatments **100**, **200**). The motorized window treatment **600** may also comprise one or more mounting brackets for mounting the motorized window treatment **600** to a structure, such as a first mounting bracket **630** (e.g., the first mounting bracket **130**A, **230**A, **330**) for supporting the motor drive unit **690** and a second mounting bracket (e.g., the second mounting bracket **130**B, **230**B) for supporting the idler. The first mounting bracket **630** and the second mounting bracket may be made from a conductive material, such as aluminum or other suitable metal. In addition, the first mounting bracket **630** and the second mounting bracket may be made from a non-conductive material, such as plastic. For example, the first mounting bracket **630** may be identical to the first mounting bracket **330** shown in FIGS. **4-9**. The motorized window treatment **600** may be adjusted between an operating position (e.g., as shown in FIGS. **1**, **4**, and **5**) to an extended position (e.g., as shown in FIGS. **2** and **3**) while secured to the first mounting bracket **630** and the second mounting bracket.

The motor drive unit **690** may comprise a housing **680**. The housing **680** may comprise a body **681** and a cap **650** (e.g., the cap **250** and/or the cap **350**). The body **681** may be cylindrical. The cap **650** may be configured to attach to the body **681**. The body **681** may comprise an upper portion **682**A and a lower portion **682**B. The motor drive unit **690** may be operatively coupled to the roller tube **611**, for example, via a coupler (e.g., the coupler **395**). The coupler may be an output gear that transfers rotation of a motor (e.g., the motor **396**) to the roller tube **611**. The motorized window treatment **600** (e.g., the motor drive unit **390**) may include a bearing assembly **620**, that may be captured between the roller tube **611** and the mounting bracket **630**. For example, the bearing assembly **620** may be identical to the bearing assembly **320**. The bearing assembly **620** may be made of non-metallic (e.g., plastic) sleeve bearings. The bearing assembly **620** may be captured between the roller tube **611** and the mounting bracket **630**, such that the bearing assembly **620** may be located in a gap **628** (e.g., a longitudinal gap) between the roller tube **611** and the mounting bracket **630**. The components of the motorized window treatment **600** in and/or adjacent to the gap **628** may be non-conductive such that radio-frequency field disruption and/or shielding is minimized. For example, the motorized window treatment **600** may not include conductive (e.g., metal) components in an area radially surrounding the gap **628**.

The motor drive unit **690** may include a battery holder **670** (e.g., the battery holder **270**, **370**). The battery holder **670** and the cap **650** may keep the batteries **660** fixed in place securely while the batteries **660** are providing power to the motor drive unit **690** and/or the cap **650**. The battery holder **670** may be configured to clamp the batteries **660** together such that the batteries **660** can be removed from the motorized window treatment **600** at the same time (e.g., together). The battery holder **670** may be received in a motor drive unit cavity **689** of the motor drive unit **690**. The batteries **660** may be configured to be removed from the motor drive unit **690**, for example, while the housing **680** of the motor drive unit **690** remains engaged with the first mounting bracket **630** and the second mounting bracket. That is, the batteries **660** may be configured to be removed from the motor drive unit **690** when the motorized window treatment **300** is in the pivoted position.

The cap **650** may be configured to cover an end of the motor drive unit cavity **689**. For example, the cap **650** may be received (e.g., at least partially) within the motor drive unit cavity **689**. The cap **650** may comprise a first portion **651** and a second portion **653**. The cap **650** may include a button **652** (e.g., formed as part of the first portion **651** of the cap **650**), a control interface printed circuit board **654** (e.g., that is housed between the first portion **651** and the second portion **653**), and an electrical contact **656** (e.g., a conductive pad) electrically coupled to the control interface printed circuit board **654**. For example, the electrical contact **656** may be a positive electrical contact. Alternatively, the electrical contact **656** may be a negative electrical contact. The cap **650** may include a switch **655** (e.g., a mechanical tactile switch) mounted to the control interface printed circuit board **654** and configured to be actuated in response to actuations of the button **652**. The button **652** may be illuminated by a light-emitting diode (LED) **658** mounted to the control interface printed circuit board **654**. The motor drive unit **690** may also comprise a wireless communication circuit (e.g., the wireless communication circuit **399**) mounted to the interface printed circuit board **654**.

As with the motor drive unit **390** shown in FIG. **4**, the motor drive unit **690** may include a motor drive printed circuit board (e.g., the motor drive printed circuit board **392**), an intermediate storage device (e.g., the intermediate storage device **394**), and a gear assembly (e.g., the gear assembly **398**). The batteries **660** may be located between the cap **650** and the motor drive printed circuit board of the motor drive unit **690**. The motor drive printed circuit board may have mounted thereto a motor drive circuit for driving the motor and a control circuit for controlling the motor drive circuit mounted to the motor drive printed circuit board. The motor drive unit **690** may include a spring (e.g., the spring **384**) configured to abut and apply a force to one of the batteries **660**, for example, such that the batteries **660** remain in contact with one another while installed within the motor drive unit cavity **689** (e.g., to maintain electrical connection of the batteries **660** with the spring and the electrical contact **656** of the cap **650**). The spring may be electrically coupled to the motor drive printed circuit board and may be a negative electrical contact. Alternatively, the spring may be a positive electrical contact.

The button **652** may be configured to enable a user to change one or more settings of the motorized window treatment **600**. For example, the button **652** may be configured to change one or more settings of the control interface printed circuit board **654** and/or the motor drive printed circuit board. The button **652** may be configured to enable a user to pair the motorized window treatment **600** with a remote control device to allow for wireless communication between the remote control device and the wireless communication circuit mounted to the interface printed circuit board **654**. The button **652** may be configured to provide a status indication to a user. For example, the control button **652** may be configured to flash and/or change colors to provide the status indication to the user. The button **652** may be configured to indicate (e.g., via the status indication) whether the motor drive unit **690** is in a programming mode.

The control interface printed circuit board **654** and the motor drive printed circuit board may be electrically connected. For example, the motorized window treatment **600** may include a ribbon cable **686**. The ribbon cable **686** may be attached to a connector **688** mounted to the control interface printed circuit board **654** and a similar connector mounted to the motor drive printed circuit board. The ribbon cable **686** may be configured to electrically connect the control interface printed circuit board **654** and the motor drive printed circuit board. The ribbon cable **686** may terminate at the control interface printed circuit board **654** and the motor printed circuit board. For example, the ribbon cable **686** may extend within the motor drive unit cavity **689**. The ribbon cable **686** may include electrical conductors for providing power from the batteries **660** to the control interface printed circuit board **654** and/or the motor drive printed circuit board. The ribbon cable **686** may include electrical conductors for conducting control signals (e.g., for transmitting one or more messages) between the control interface printed circuit board **654** and the motor drive printed circuit board. For example, the ribbon cable **686** may be configured to conduct power and/or control signals between the control interface printed circuit board **654** and the motor drive printed circuit board.

The motor drive unit **690** may further comprise an antenna **700**. For example, the antenna **700** may comprise an insulated electrical conductor, such as a 22-gauge stranded electrical wire with a polyvinyl chloride (PVC) coating. The antenna **700** may be wrapped around (e.g., wound about) the housing **680**. The antenna **700** may be located in a channel **711** (e.g., a spiral-shaped channel) that extends around the cap **650** of the housing **630** of the motor drive unit **690** (e.g., about the periphery of the cap **650**), such that the antenna **700** may be wrapped around (e.g., wound about) the cap **650**. For example, the channel **711** may be similar in shape as the spiral-shaped channel **511** that extends around the housing **380'** of the motor drive unit **390'** (e.g., as shown in FIG. **11**). The channel **711** may be a continuous spiral that moves farther and farther away, in the longitudinal direction L, from the roller tube **611**. Alternatively, the channel **711** may comprise at least two peripheral channels that extend parallel to each other around the circumference of the cap **650** (e.g., similar to the first and second peripheral channels **411**, **412**). The at least two peripheral channels may be joined together at a recess (e.g., similar to the recess **414**) such that the electrical conductor of the antenna **700** is configured to pass from one peripheral channel **711** to another via the recess. The antenna **700** may be located within the channel **711**, such that the antenna **700** may be wrapped around the cap **650**. The electrical wire of the antenna **700** may be held in the channel **711** by tabs (e.g., such as the tabs **419**). A distance (e.g., in the longitudinal direction L) between windings of the antenna **700** may be configured to prevent the antenna **700** from coupling to itself. The antenna **700** may comprise an insulated electrical conductor, for example, such as a 22-gauge stranded electrical wire with a polyvinyl chloride (PVC) coating.

As shown in FIG. **15**, the antenna **700** may be wrapped around the cap **650** of the housing **680** of the motor drive unit **690** adjacent to an end **613** of the roller tube **611**. For example, the antenna **700** may be wrapped around the cap **650** of the housing **680** adjacent to the gap **628** between the roller tube **611** and the mounting bracket **630**. The antenna **700** may transmit and/or receive RF signals through the gap **628**. In addition, the antenna **700** may be configured to couple (e.g., capacitively couple) to the roller tube **611** (e.g., when the roller tube **611** is made of a conductive material), such that standing waves are generated on a surface of the roller tube **611**, which may increase the amount of RF signals transmitted and/or received by the antenna **700**.

As shown in FIG. **15**, the channel **711** may be formed in an inner surface of the second portion **653** of the cap **650**. In addition, the channel **711** may be formed in an outer surface of the second portion **653** and/or an inner or outer surface of the first portion **651**. Rather than the channel **711** being a single spiral-shaped channel, the cap **650** of the housing **680** may comprise one or more channels, such as parallel peripheral (e.g., circumferential) channels (e.g., as with the first and second peripheral channels **411**, **412** in the housing **380** of the motor drive unit **390** as shown in FIG. **8**).

The antenna **700** may be mechanically and electrically coupled to the control interface printed circuit board **654** for electrically coupling to the wireless communication circuit on the control interface printed circuit board **654**. The control interface printed circuit board **654** may have a matching network circuit (e.g., similar to the matching network circuit **426**) mounted thereto. The matching network circuit may be electrically coupled between the antenna **700** and the wireless communication circuit mounted to the control interface printed circuit board **654**. The matching network circuit on the control interface printed circuit board **654** may be configured to optimize the performance of the antenna **700**. For example, the matching network circuit may be configured to match an impedance of the antenna **700** to an impedance of the wireless communication circuit on the control interface printed circuit board **654** to obtain a maximum transfer of power between the antenna **700** and the wireless communication circuit. The matching network circuit may include, for example, an inductor-capacitor (LC) filter. The wireless communication circuit on the control interface printed circuit board **654** may be electrically coupled to a motor control circuit on the motor control printed circuit board via the ribbon cable **686**. Alternatively, the wireless communication circuit may be mounted to the motor drive printed circuit board, and the matching network circuit on the control interface printed circuit board **654** may be electrically coupled to the wireless communication circuit via a coaxial cable (e.g., the coaxial cable **430**). The coaxial cable may extend through a coaxial cable channel (e.g., similar to coaxial cable channel **432** formed in the inner surface **424** of the upper portion **382**A of the housing **380**) in the longitudinal direction L of the motorized window treatment **600**.

When the wireless communication circuit on the control interface printed circuit board **654** drives the antenna **700** with a signal, the antenna **700** may emit electromagnetic waves (e.g., radio-frequency signals). As previously mentioned, the antenna **700** may be wrapped around the cap **650** of the housing **680** of the motor drive unit **690** adjacent to the gap **628** between the roller tube **611** and the mounting bracket **630**. When the antenna **700** is emitting electromagnetic waves, the electromagnetic waves may be coupled to the roller tube **611** (e.g., which may be made of a conductive material, such as aluminum), which may result in current flow on the surface of the roller tube **611** (e.g., standing waves). As a result, the roller tube **311** may re-radiate the electromagnetic waves emitted by the antenna **700**.

While the antenna **700** is shown in FIG. **15** being wrapped around the motor drive units **390** one or more times, the antenna **700** may be wrapped around the motor drive unit **390** in other manners and/or shapes. For example, the one or more channels in which the antenna **700** is located may be formed on the inner surface **424** of the upper portion **382**A of the housing **380**. In addition, the antenna **700** may be internal to the housing **380** of the motor drive unit **390** (e.g., extending through one or more tunnels in the housing **380**). Alternatively, the one or more channels in which the antenna **700** is located may be formed in the bearing assembly **320** (e.g., such as the stationary bearing **322**).

FIG. **16** is a block diagram of an example motor drive unit **800** (e.g., the motor drive unit **390** shown in FIGS. **4-9**, the motor drive unit **390'** shown in FIGS. **12-14** and/or the motor drive unit **690** shown in FIG. **15**) of a motorized window treatment (e.g., such as the motorized window treatment **100** shown in FIGS. **1** and **2**, the motorized window treatment **200** shown in FIG. **3**, the motorized window treatment **300** shown in FIGS. **4** and **5**, the motorized window treatment **300'** shown in FIGS. **12-14** and/or the motorized window treatment **600** shown in FIG. **15**). The motor drive unit **800** may comprise a motor **810** (e.g., a direct-current (DC) motor) that may be coupled for raising and lowering a covering material. For example, the motor **810** may be coupled to a roller tube (e.g., roller tube **311** shown in FIGS. **4-5**) of the motorized window treatment for rotating the roller tube for raising and lowering a flexible material (e.g., a shade fabric). The motor drive unit **800** may comprise a load control circuit, such as a motor drive circuit **820** (e.g., an H-bridge drive circuit) that may generate a pulse-width modulated (PWM) voltage V_{PWM} for driving the motor **810** (e.g., to move the covering material between a fully-raised position and a fully-lowered position). In addition, the control circuit **830** may be configured to generate a direction signal for controlling the direction of rotation of the motor **810**.

The motor drive unit **800** may comprise a control circuit **830** for controlling the operation of the motor **810**. The control circuit **830** may comprise, for example, a microprocessor, a programmable logic device (PLD), a microcontroller, an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any suitable processing device or control circuit. The control circuit **830** may be configured to generate a drive signal V_{DRV} for controlling the motor drive circuit **820** to control the rotational speed of the motor **810** (e.g. the motor drive circuit **820** receives the drive signal V_{DRV} and controls, for example, an H-bridge circuit with appropriate PWM signals in response to the drive signal). In examples, the drive signal V_{DRV} may comprise a pulse-width modulated signal, and the rotational speed of the motor **810** may be dependent upon a duty cycle of the pulse-width modulated signal. In examples, the control circuit **830** may directly control the motor **810** (e.g. in a configuration with no separate motor drive circuit **820**). For example, the control circuit may generate two PWM signals for controlling the duty cycle and the polarity (e.g. controlling the speed and direction) of the motor **810**. In addition, the control circuit **830** may be configured to generate a direction signal V_{DIR} for controlling the motor drive circuit **820** to control the direction of rotation of the motor **810**. The control circuit **830** may be configured to control the motor **810** to adjust a present position P_{PRES} of the covering material of the motorized window treatment between a fully-raised position $P_{FULLY\text{-}RAISED}$ and a fully-lowered position $P_{FULL\text{-}LOWERED}$.

The motor drive unit **800** may include a rotational sensing circuit **840**, e.g., a magnetic sensing circuit, such as a Hall effect sensor (HES) circuit, which may be configured to generate two signals V_{S1}, V_{S2} (e.g., Hall effect sensor signals) that may indicate the rotational position and direction of rotation of the motor **810**. The rotational sensing circuit **840** (e.g., HES circuit) may comprise two internal sensing circuits for generating the respective signals V_{S1}, V_{S2} (e.g., HES signals) in response to a magnet that may be attached to a drive shaft of the motor **810**. The magnet may be a circular magnet having alternating north and south pole regions, for example. For example, the magnet may have two opposing north poles and two opposing south poles, such that each sensing circuit of the rotational sensing circuit **840** is passed by two north poles and two south poles during a full rotation of the drive shaft of the motor **810**. Each sensing circuit of the rotational sensing circuit **840** may drive the respective signal V_{S1}, V_{S2} to a high state when the sensing circuit is near a north pole of the magnet and to a low state when the sensing circuit is near a south pole. The control circuit **830** may be configured to determine that the motor **810** is rotating in response to the signals V_{S1}, V_{S2} generated by the rotational sensing circuit **840**. In addition, the control circuit **830** may be configured to determine the rotational position and direction of rotation of the motor **810** in response to the signals V_{S1}, V_{S2}.

The motor drive unit **800** may include a communication circuit **842** (e.g., such as the control interface printed circuit board **354** shown in FIGS. **4** and **5**) that may allow the control circuit **830** to transmit and receive communication signals, e.g., wired communication signals and/or wireless communication signals, such as radio-frequency (RF) signals. For example, the motor drive unit **800** may be configured to communicate messages (e.g., digital messages) with external control devices (e.g., other motor drive units) via the communication circuit **842** (e.g., via wireless signals, such as RF signals). The wireless communication circuit **842** may be coupled to an antenna **845** (e.g., the antenna **400** and/or the antenna **500**) via a matching network **849** (e.g., the matching network **426** mounted to the coupling printed circuit board **420** and/or the matching network **526** mounted to the coupling printed circuit board **520**). The matching network **849** may be configured to optimize the performance of the antenna **845**. For example, the matching network **849** may include a filter (e.g., an inductor-capacitor (LC) filter) that is configured to match an impedance of the antenna **845** to an impedance of the wireless communication circuit **842** to obtain a maximum transfer of power between the antenna **845** and the communication circuit **842**. The communication circuit **842** and/or the antenna **845** may be communicatively coupled (e.g., electrically connected) to the control circuit **830**.

The motor drive unit **800** may communicate with one or more input devices, e.g., such as a remote control device, an occupancy sensor, a daylight sensor, and/or a shadow sensor. The remote control device, the occupancy sensor, the daylight sensor, and/or the shadow sensor may be wireless control devices (e.g., RF transmitters) configured to transmit messages to the motor drive unit **800** via the RF signals. For example, the remote control device may be configured to transmit digital messages via the RF signals in response to an actuation of one or more buttons of the remote control device. The occupancy sensor may be configured to transmit messages via the RF signals in response to detection of occupancy and/or vacancy conditions in the space in which the motorized window treatment is installed. The daylight sensor may be configured to transmit digital messages via RF signals in response to a measured amount of light inside of the space in which the motorized window treatment is installed. The shadow sensor may be configured to transmit messages via the RF signals in response to detection of a glare condition outside the space in which the motorized window treatment is installed.

The motorized window treatment may be configured to control the covering material according to a timeclock schedule. The timeclock schedule may be stored in the memory. The timeclock schedule may be defined by a user (e.g., a system administrated through a programming mode). The timeclock schedule may include a number of timeclock events. The timeclock events may have an event time and a corresponding command or preset. The motorized window treatment may be configured to keep track of the present time and/or day. The motorized window treatment may transmit the appropriate command or preset at the respective event time of each timeclock event.

The motor drive unit **800** may further comprise a user interface **844** having one or more actuators (e.g., mechanical switches) that allow a user to provide inputs to the control circuit **830** during setup and configuration of the motorized window treatment (e.g., in response to actuations of one or more buttons (e.g., the control button **152** shown in FIG. **1**). The control circuit **830** may be configured to control the motor **810** to control the movement of the covering material in response to a shade movement command received from the communication signals received via the communication circuit **842** or the user inputs from the buttons of the user interface **844**. The control circuit **830** may be configured to enable (e.g., via the control button **152** and/or the user interface **844**) a user to pair the motorized window treatment with a remote control device and/or other external devices to allow for wireless communication between the remote control device and/or other external devices and the communication circuit **842** (e.g., an RF transceiver). The user interface **844** (e.g., the control button **152**) may be configured to provide a status indication to a user. For example, user interface **844** (e.g. the control button **152**) may be configured to flash and/or change colors to provide the status indication to the user. The status indication may indicate when the motorized window treatment is in a programming mode. The user interface **844** may also comprise a visual display, e.g., one or more light-emitting diodes (LEDs), which may be illuminated by the control circuit **830** to provide feedback to the user of the motorized window treatment system.

The motor drive unit **800** may comprise a memory (not shown) configured to store the present position P_{PRES} of the covering material and/or the limits (e.g., the fully-raised position $P_{FULLY\text{-}RAISED}$ and the fully-lowered position $P_{FULLY\text{-}LOWERED}$), association information for associations with other devices and/or instructions for controlling the motorized window treatment. The memory may be implemented as an external integrated circuit (IC) or as an internal circuit of the control circuit **830**.

The motor drive unit **800** may comprise a compartment **864** (e.g., which may be an example of the battery compartment **211** of the window treatment **200** shown in FIG. **3**) that is configured to receive a DC power source. In some examples, the compartment **864** may be internal to the motor drive unit **800**. In other examples, the compartment **864** may be external to the motor drive unit **800**. In the example shown in FIG. **3**, the DC power source is one or more batteries **860**. In addition, alternate DC power sources, such as a solar cell (e.g., a photovoltaic cell), an ultrasonic energy source, and/or a radio-frequency (RF) energy source, may be coupled in parallel with the one or more batteries **860**, or in some examples be used as an alternative to the batteries **860**. The alternate DC power source may be used to perform the same and/or similar functions as the one or more batteries **860**. In this example, the compartment **864** may be configured to receive one or more batteries **860** (e.g. four "D" batteries), such as the batteries **260**, **360** of FIGS. **3**-**5**. The batteries **860** may provide a battery voltage V_{BATT} to the motor drive unit **800**.

The motor drive unit **800** may comprise a filter circuit **870**, a current limiting circuit, such as a power converter circuit **852**, and an energy storage element **854** (e.g., an intermediate energy storage element such as the intermediate storage device **694** shown in FIG. **8**A). In some examples, the motor drive unit **800** may include a second power converter, such as a boost converter circuit **858**. Also, in some examples, the second power converter may be omitted from the motor drive circuit **800**. The energy storage element **854** may comprise any combination of one or more super capacitors, one or more rechargeable batteries, and/or other suitable energy storage devices.

The filter circuit **870** may receive the battery voltage V_{BATT}. The power converter circuit **852** may draw a battery current I_{BATT} from the batteries **860** through the filter circuit **870**. The filter circuit **870** may filter high and/or low frequency components of the battery current I_{BATT}. In some examples, the filter circuit **870** may be a low-pass filter. Also, in some examples, the filter circuit **870** may be omitted from the motor drive circuit **800**.

The power converter circuit **852** may be configured to limit the current drawn from the batteries **860** (e.g. allowing a small constant current to flow from the batteries **860**). The power converter circuit **852** may receive the battery voltage V_{BATT} via the filter circuit **870**. In some examples, the power converter circuit **852** may comprise a step-down power converter, such as a buck converter. The power converter circuit **852** may be configured to charge the energy storage element **854** from the battery voltage V_{BATT} to produce a storage voltage V_S across the energy storage element **854** (e.g., approximately 3.5 volts). The motor drive circuit **820** may draw energy from the energy storage element **854** (e.g., via the boost converter circuit **858**) to drive the motor **810**. As such, the power converter circuit **852** may be configured to limit the current drawn from the batteries **860**, for example, by producing a storage voltage V_S and driving the motor **810** using the storage voltage V_S stored across the energy storage element **854**. In most cases, for instance, the motor drive circuit **820** may drive the motor **810** by drawing current from the energy storage element **854** and not drawing any current directly from the batteries **860**. Further, it should be appreciated that, in some examples, the power converter circuit **852** may be omitted for another current limiting circuit, such as in instances where the battery voltage V_{BATT} is the same as the storage voltage V_S and power conversion (e.g., a step-up or step-down) is not needed to drive the motor **810**.

The motor drive unit **800** may be configured to control when and how the energy storage element **854** charges from the batteries **860**. The control circuit **830** may control when and how the energy storage element **854** charges from the batteries **860** based on the storage voltage V_S of the energy storage element **854**, such as when the storage voltage V_S of the energy storage element **854** falls below a low-side threshold value (e.g., approximately 2.8 volts). For example, the control circuit **830** may be configured to receive a scaled storage voltage V_{SS} via a scaling circuit **856** (e.g., a resistive divider circuit). The scaling circuit **856** may receive the storage voltage V_S and may generate the scaled storage voltage V_{SS}. The control circuit **830** may determine the magnitude of the storage voltage V_S of the energy storage element **854** based on the magnitude of the scaled storage voltage V_{SS}. When the control circuit **830** determines that the magnitude of the storage voltage V_S of the energy storage element **854** falls below the low-side threshold value, the control circuit **830** may control a charging enable signal V_{EN} (e.g., drive the charging enable control signal V_{EN} high) to enable the power converter circuit **852**. When the power converter circuit **852** is enabled, the power converter circuit **852** may be configured to charge the energy storage element **854** (e.g. from the batteries **860**). When the power converter circuit **852** is disabled, the power converter circuit **852** may be configured to cease charging the energy storage element **854** (e.g. from the batteries **860**).

The motor drive unit **800** may utilize the energy storage element **854** to draw a small constant current from the batteries **860** over a long period of time to extend the lifetime (e.g., and increase the total energy output) of the batteries **860**. For example, the motor drive unit **800** (e.g., the power converter circuit **852** and/or the motor drive circuit **820**) may limit the current drawn by the power converter circuit **852**. The motor drive unit **800** may draw current from the batteries **860** that is less than the limit, but not more.

When enabled, the power converter circuit **852** may be configured to conduct an average current I_{AVE} (e.g., having a magnitude of approximately 15 milliamps) from the batteries **860**. The magnitude of the average current I_{AVE} may be much smaller than a magnitude of a drive current required by the motor drive circuit **820** to rotate the motor **810**. When the motor drive circuit **820** is driving the motor **810**, the magnitude of the storage voltage V_S of the energy storage element **854** may decrease with respect to time. When the motor drive circuit **820** is not driving the motor **810** and the power converter circuit **852** is charging the energy storage element **854**, the magnitude of the storage voltage V_S may increase (e.g., slowly increase). When the storage voltage V_S of the energy storage element **854** falls below a low-side threshold value (e.g. approximately 2.8V), the control circuit **830** may enable the power converter circuit **852** to begin charging the energy storage element **854**. The storage voltage V_S may fall below the low-side threshold value after powering movements of the covering material, powering low-voltage components, and/or due to leakage currents over time. When the storage voltage V_S of the energy storage element **854** rises above a high-side threshold value (e.g., approximately 3.5 volts), the control circuit **830** may cease driving the charging enable signal V_{EN} high to disable the power converter circuit **852** and stop the charging of the energy storage element **854** from the batteries **860**.

The motor drive unit **800** may further comprise the boost converter circuit **858** that receives the storage voltage V_S and generates a motor voltage V_{MOTOR} (e.g., approximately 5 volts) for powering the motor **810**. The motor voltage V_{MOTOR} may be larger than the storage voltage V_S. In some examples, a switch (e.g., a single pole double throw switch) may connect the batteries **860** and the energy storage element **854** to the boost converter **858** (e.g., if the required motor voltage level exceeds the present battery voltage V_{BAT}). When the control circuit **830** controls the motor drive circuit **820** to rotate the motor **810**, the boost converter circuit **858** may conduct current from the energy storage element **854** to generate the motor voltage V_{MOTOR}. As noted above, in some examples, the motor drive unit **800** may not include the boost converter circuit **858**, for example, based on the voltage requirements of the motor **810**.

The motor drive unit **800** may also comprise a controllable switching circuit **862** coupled between the batteries **860** and the motor drive circuit **820**. The control circuit **830** may generate a switch control signal V_{SW} for rendering the controllable switching circuit **862** conductive and non-conductive. The control circuit **830** may be configured to render the controllable switching circuit **862** conductive to bypass the filter circuit **870**, the power converter circuit **852**, the energy storage element **854**, and/or the boost converter circuit **858** to allow the motor drive circuit **820** to draw current directly from the batteries (e.g., when the energy storage element **854** is depleted). For example, the control circuit **830** may render the controllable switching circuit **862** conductive when the control circuit **830** determines that the magnitude of the storage voltage V_S of the energy storage element **854** (e.g., based on the magnitude of the scaled storage voltage V_{SS}) is depleted below a threshold and the control circuit **830** has received an input or command to operate the motor **810** and, for example, does not have enough energy to complete a movement or an amount of movement of the covering material). For example, the control circuit may determine if the energy storage element **854** has enough energy to complete a movement or an amount of movement of the covering material by comparing a present storage level of the energy storage element **854** (e.g., the storage voltage V_S) to a threshold. The threshold may indicate a storage level sufficient to complete a full movement of the covering material from the fully-lowered position $P_{FULLY\text{-}LOWERED}$ to the fully-raised position $P_{FULLY\text{-}RAISED}$ (e.g., a fixed threshold). The threshold may be constant or may vary, for example, depending on the amount of movement of the covering material required by the received command, such that the threshold (e.g., a variable threshold) may indicate a storage level sufficient to complete the movement required by the received command.

If the energy storage element **854** is not sufficiently charged (e.g., does not have enough energy to move the covering material), the control circuit **830** may close the controllable switching circuit **862** at to allow the electrical load (e.g., the motor) to draw current directly from the batteries **860**. Closing the controllable switching circuit **862** may bypass the energy storage element **854**, such that the stored energy of the energy storage element **854** is not used for driving the motor **810** to move the covering material.

The control circuit **830** may be configured to determine when one or more of the batteries **860** are not installed in the compartment **864** when in the operating position. For example, the control circuit **830** may be configured to determine that one or more of the batteries **860** are missing when the magnitude of the battery voltage V_{BATT} drops to approximately zero volts (e.g., there is an open circuit between the battery contacts). The control circuit **830** may be configured to determine the magnitude of the battery voltage V_{BATT} in response to a scaled battery voltage V_{BATT-S} received via a scaling circuit **866** (e.g., a resistive divider circuit). The scaling circuit **866** may receive the battery voltage V_{BATT} and may generate the scaled battery voltage V_{BATT-S}. The control circuit **830** may be configured to disable (e.g., automatically disable) the operation of the motor **810** of the motor drive unit **800** in response to the scaled battery voltage V_{BATT-S}, such that the covering material cannot be raised or lowered when one or more of the batteries **860** are not installed in the battery compartment **864**, which may prevent depletion of the intermediate storage element **854**. The control circuit **830** may be configured to enable the operation of the motor **810** in response to the scaled battery voltage V_{BATT-S} when all of the batteries **860** are installed.

The motor drive unit **800** may comprise a power supply **880** (e.g., a low-voltage power supply). The power supply **880** may receive the battery voltage V_{BATT}. The power supply **880** may be configured to produce a low-voltage supply voltage V_{CC} (e.g., approximately 3.3 volts) for powering low-voltage circuitry of the motor drive unit **800**, such as the user interface **844**, the communication circuit **842**, and the control circuit **830**. Further, in some examples, the power supply **880** may be omitted from the motor drive unit **800** (e.g. if the low-voltage circuitry of the motor drive unit **800** is able to be powered directly from the storage voltage V_S). Additionally or alternatively, the motor drive unit **800** may comprise a power supply (not shown) that may receive the storage voltage V_S and generate the low voltage V_{CC} (e.g., approximately 3.3 V) for powering the control circuit **830** and other low-voltage circuitry of the motor drive unit **800**, e.g., the user interface **844**, the communication circuit **842**, and the control circuit **830**.

The invention claimed is:

1. A motorized window treatment comprising:

a roller tube configured to windingly receive a flexible material and to be rotated to raise and lower the flexible material;

a motor drive unit received within a cavity of the roller tube, the motor drive unit comprising:

a motor configured to rotate the roller tube;

a housing configured to house the motor, the housing comprising at least one channel formed in a surface of the housing; and

an antenna comprising an electrical conductor;

at least one mounting bracket configured to support the roller tube such that the roller tube can rotate with respect to the at least one mounting bracket; and

a bearing assembly coupled to the roller tube and located between the roller tube and the mounting bracket,

wherein a gap is defined between the roller tube and the mounting bracket, and wherein the electrical conductor of the antenna is wrapped around the housing of the motor drive unit adjacent to the gap between the roller tube and the mounting bracket and within an area defined by a rotatable portion of the bearing assembly, wherein the electrical conductor of the antenna is configured to be received within the at least one channel when wrapped around the housing.

2. The motorized window treatment of claim **1**, wherein the motor drive unit comprises a wireless communication circuit electrically coupled to the antenna for transmitting and receiving wireless signals.

3. The motorized window treatment of claim **1**, wherein the at least one channel comprises at least two peripheral channels that extend parallel to each other around the circumference of the housing in an outer surface of the housing.

4. The motorized window treatment of claim **3**, wherein the at least two peripheral channels are joined together at a recess and the electrical conductor of the antenna is configured to pass from one peripheral channel to another via the recess.

5. The motorized window treatment of claim **1**, wherein the at least one channel comprises a single spiral-shaped channel.

6. The motorized window treatment of claim **5**, wherein the spiral-shaped channel is configured such that the antenna moves away from the roller tube in the longitudinal direction as the antenna wraps around the housing.

7. The motorized window treatment of claim **2**, wherein the motor drive unit comprises a motor drive printed circuit board on which drive circuitry for controlling the motor is mounted.

8. The motorized window treatment of claim **7**, wherein the motor drive unit further comprises a battery compartment for receiving one or more batteries for powering the drive circuitry on the motor drive printed circuit board and the wireless communication circuit, the housing comprising a cap for covering an end of the battery compartment, the battery compartment located between the cap and the motor drive printed circuit board.

9. The motorized window treatment of claim **8**, wherein the wireless communication circuit is located inside the cap.

10. The motorized window treatment of claim **8**, further comprising a matching network circuit that is coupled to the motor drive printed circuit board, wherein the matching network circuit is located inside the cap.

11. The motorized window treatment of claim **9**, wherein the wireless communication circuit is coupled to the motor drive printed circuit board via a ribbon cable.

12. The motorized window treatment of claim **7**, wherein the motor drive unit comprises a coupling printed circuit board located near the gap between the roller tube and the mounting bracket and having a matching network circuit mounted thereto, and wherein the antenna is electrically coupled to the matching network circuit on the coupling printed circuit board.

13. The motorized window treatment of claim **12**, wherein the wireless communication circuit is mounted to the motor drive printed circuit board and electrically connected to the matching network circuit on the coupling printed circuit board by a coaxial cable.

14. The motorized window treatment of claim **1**, wherein the motor drive unit comprises a flexible printed circuit board, and wherein the antenna is formed on the flexible printed circuit board.

15. The motorized window treatment of claim **1**, wherein the bearing assembly enables the roller tube is to rotate around the motor drive unit.

16. The motorized window treatment of claim 15, wherein the bearing assembly is made of a non-conductive material.

17. The motorized window treatment of claim 16, wherein the antenna is wrapped around the motor drive unit within an area that surrounds the circumference of the motor drive unit underneath the bearing assembly.

18. The motorized window treatment of claim 1, wherein at least a portion of the antenna is aligned with the gap between the roller tube and the at least one mounting bracket.

19. The motorized window treatment of claim 18, wherein the gap between the roller tube and the at least one mounting bracket defines an area comprising non-conductive components.

20. The motorized window treatment of claim 1, wherein the roller tube is made of a conductive material, and the antenna is configured to be electromagnetically coupled to the roller tube.

21. The motorized window treatment of claim 1, wherein the roller tube and the mounting bracket are both made of conductive materials.

22. The motorized window treatment of claim 1, wherein the housing comprises a body and a cap that is configured to attach to the body.

23. The motorized window treatment of claim 22, wherein the at least one channel is defined in the body such that the antenna is wrapped around the body.

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